CIVL1118 Structural Mechanics



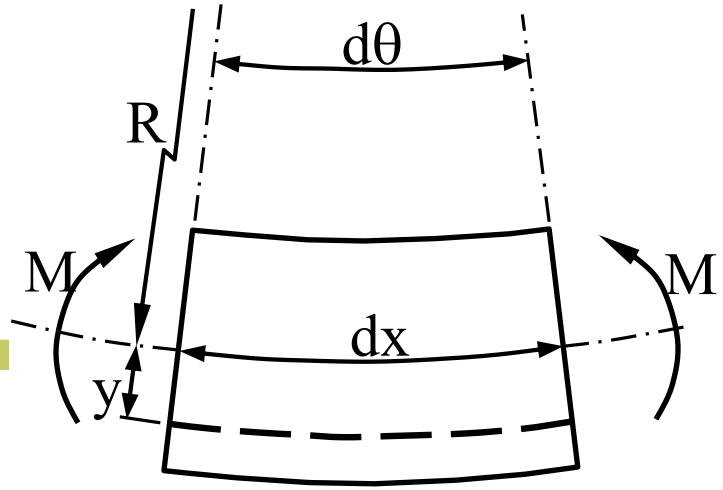
Deflection

Dr Jiaji Wang

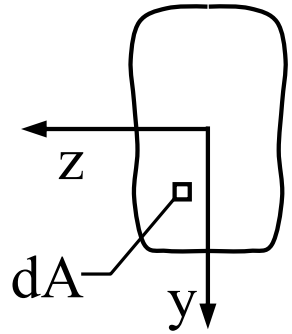
Deflection of beams

Assumptions

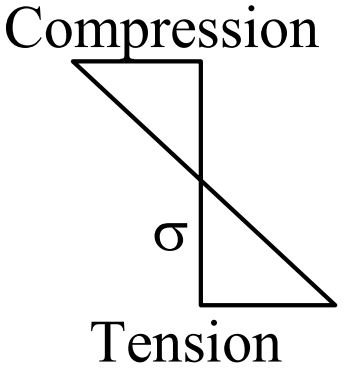
1. The beam is prismatic, initially straight and has an axial plane of symmetry which is taken as the x-y plane.
2. The material is homogeneous and obeys Hooke's Law.
3. Plane sections before bending remain plane after bending.



Beam segment under pure bending



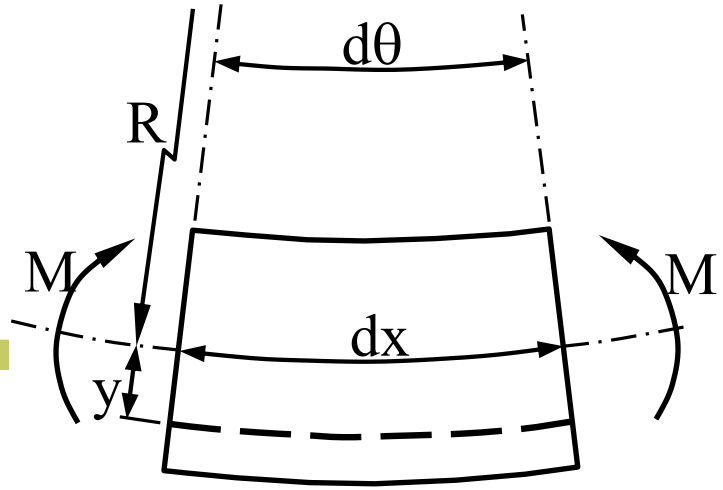
Beam section



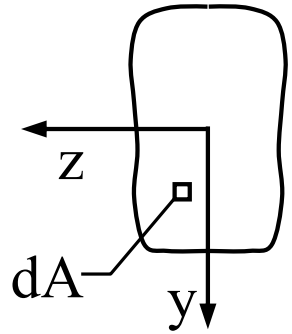
Bending stress

Beam under pure bending

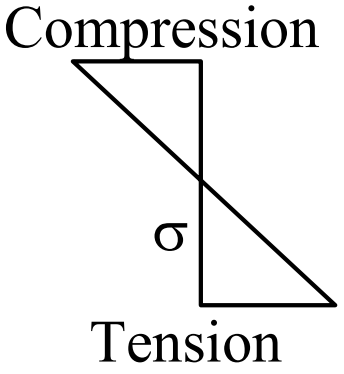
- As a result of deformation, fibres on the concave side are shortened slightly (compression), while those on the convex side are elongated slightly (tension).
- Somewhere in between, the section remains unchanged in length, i.e. *neutral surface.*
  - its intersection with any cross-section is the *neutral axis*



Beam segment under pure bending



Beam section



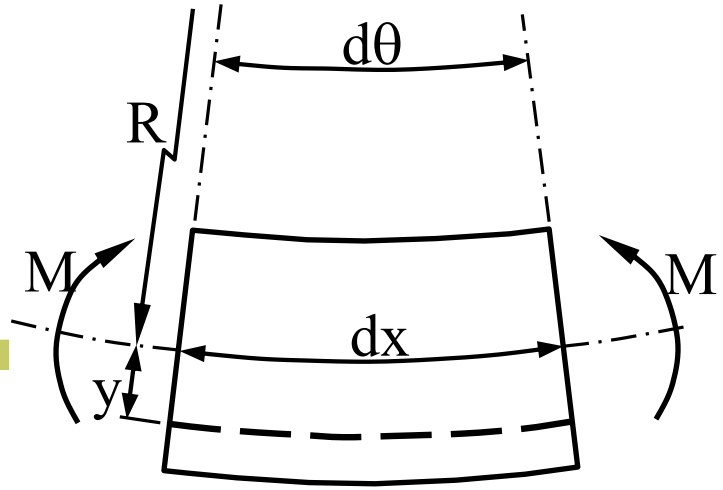
Bending stress

Beam under pure bending

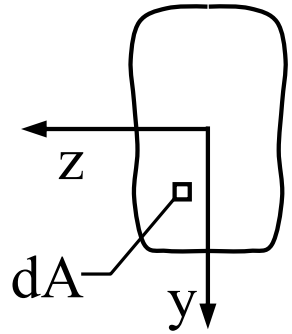
Consider two adjacent cross-sections at a small distance dx apart. After bending they intersect at o and the angle between them is denoted by dθ.

Let R = radius of curvature of the neutral surface

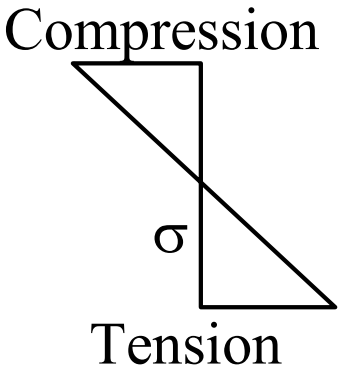
Then $d\theta = dx / R$



Beam segment under pure bending



Beam section



Bending stress

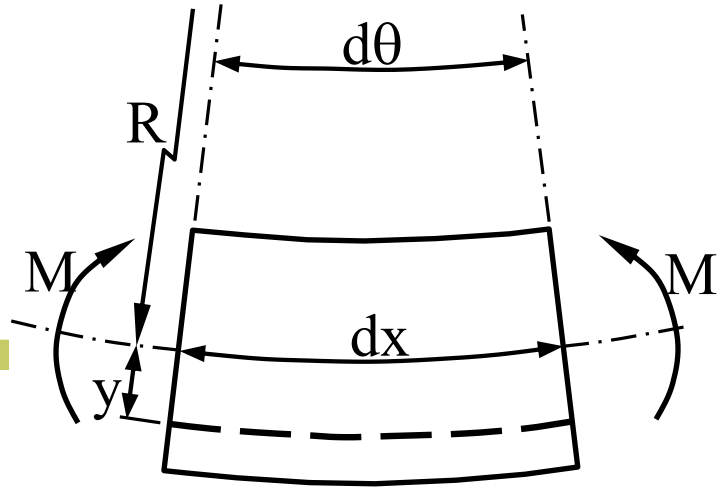
Beam under pure bending

At a depth y from the neutral axis,

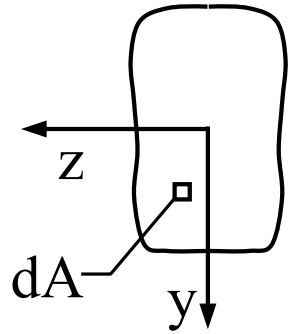
the strain is $\varepsilon = (y \cdot d\theta) / dx = y / R$

the stress is $\sigma = E \cdot \varepsilon = E \cdot y / R$

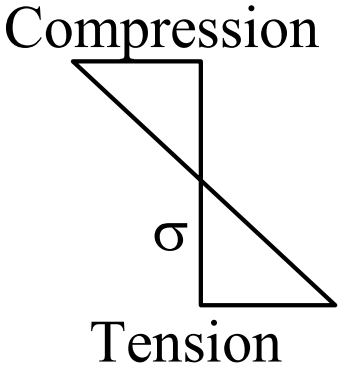
the force on a small area dA is $\sigma \cdot dA = (E/R) \cdot y \cdot dA$



Beam segment under pure bending



Beam section



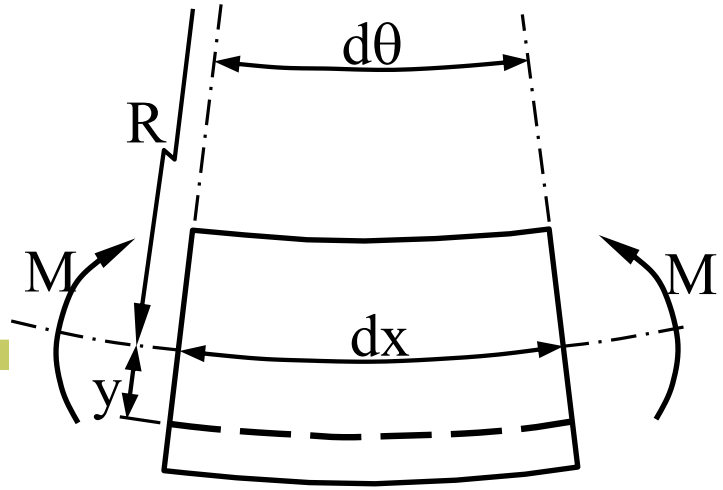
Bending stress

Beam under pure bending

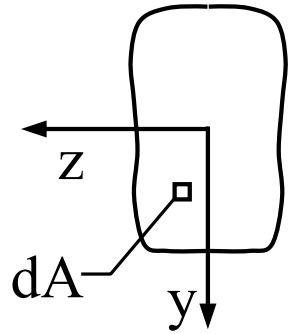
Since there must be no resultant normal force on the section under pure bending,

$$\int \sigma \, dA = (E/R) \int y \, dA = 0 \Rightarrow A\bar{y} = 0$$

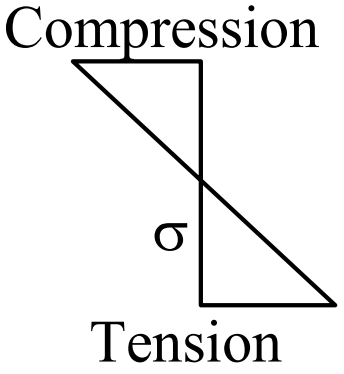
Therefore $\bar{y}$ must be equal to 0. In other words, the neutral axis of a section in pure bending passes through the centroid.



Beam segment under pure bending



Beam section



Bending stress

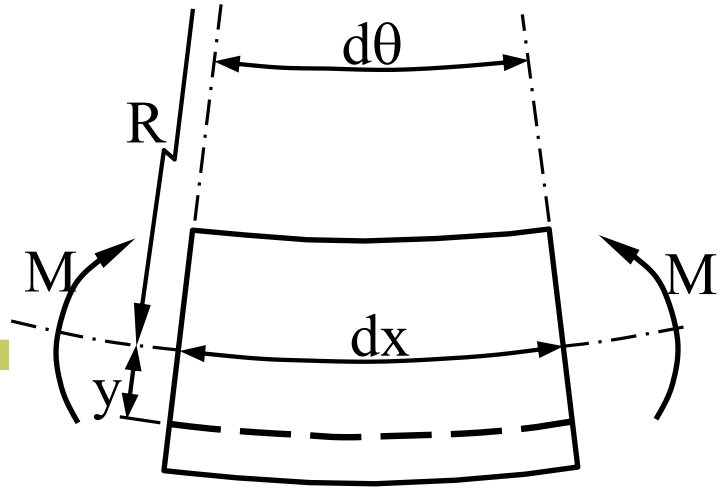
Beam under pure bending

The moment of the elemental force about the neutral axis of the section is

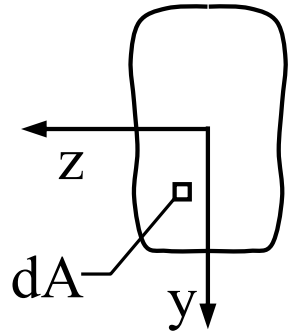
$$dM = (\sigma\, dA)\, y$$

The total moment on the section is therefore

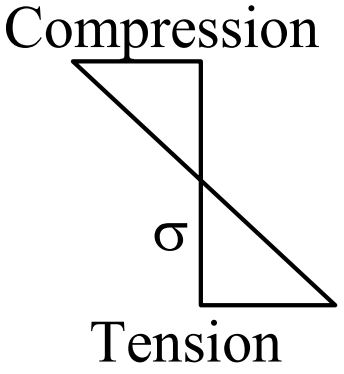
$$M = \int \sigma\, y\, dA = \int (E/R)\, y^2\, dA = (E/R) \int y^2\, dA$$



Beam segment under pure bending



Beam section



Bending stress

Beam under pure bending

The integral $\int y^2 \, dA$ is the second moment of area of the section and is denoted by I.

Therefore

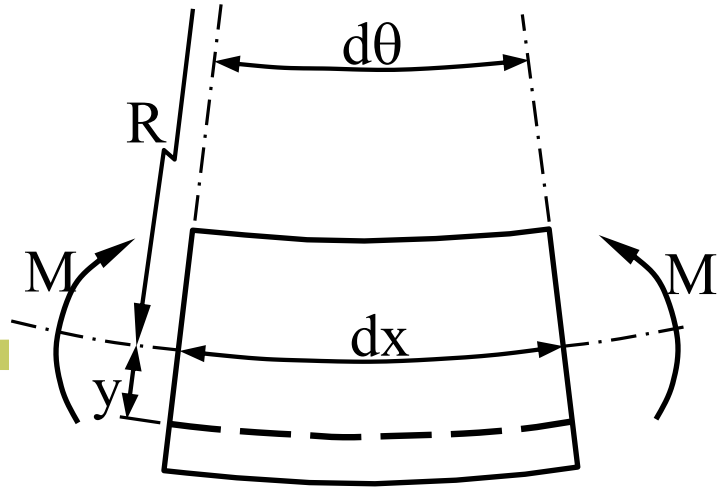
$M = (E/R)\, I$ or $1/R = M / (EI) = d\theta/dx$

Hence $\sigma = (M\, y) / I$ Note $\sigma = E \cdot y / R$

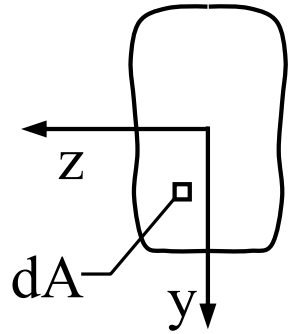
For a symmetrical prismatic beam bent by transverse loads acting in a plane of symmetry, it can be shown that the curvature of the neutral surface is given by

$$1 / R = M / EI$$

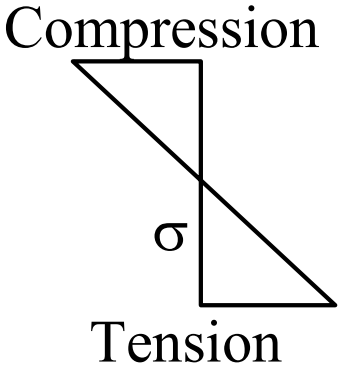
at each cross-section. Thus, the curvature $1/R$ is seen to vary along the beam as the bending moment varies.



Beam segment under pure bending



Beam section



Bending stress

Beam under pure bending

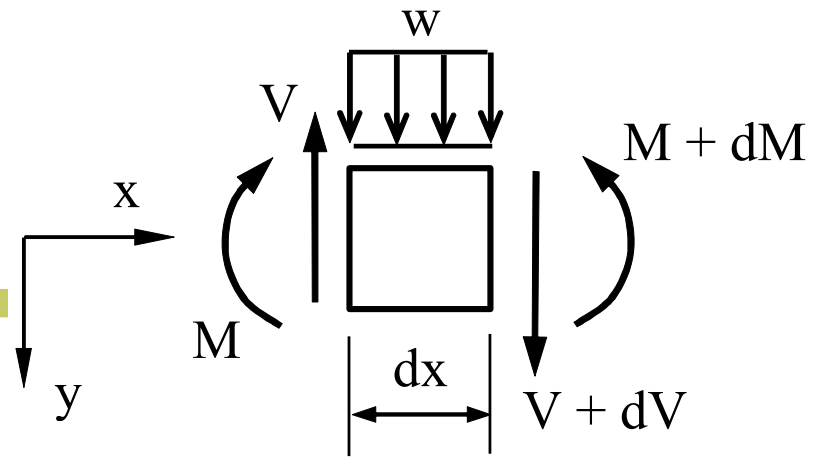
Consider any segment ds of the deflection curve (or elastic line). One has

$$ds = R\,d\theta \qquad \text{or} \qquad \frac{d\theta}{ds} = \frac{1}{R}$$

y as deflection

If the deflection is relatively small, $ds \approx dx$

$$\theta \approx \frac{dy}{dx} \qquad \frac{1}{R} \approx \frac{d^2y}{dx^2}$$

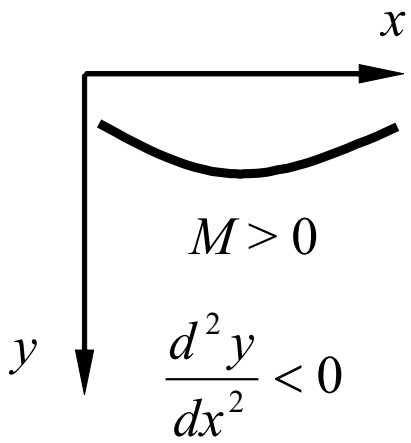


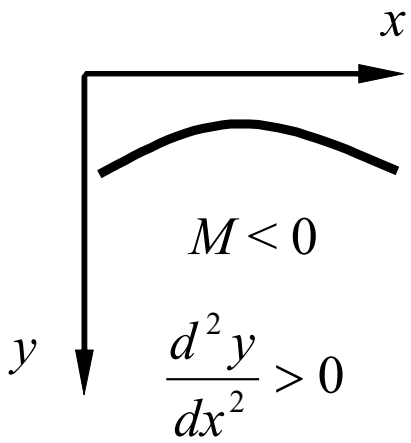
An element of a beam

Taking the sign convention and the direction of the coordinate axes into consideration, we get the differential equation of the elastic line as

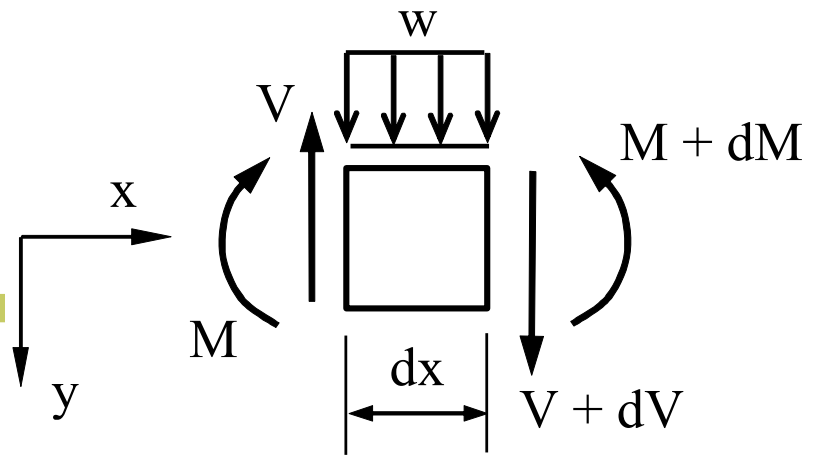
$$\frac{1}{R} = \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = -\frac{M}{EI}$$

$$EI\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = -M$$





$$EI\frac{d^2y}{dx^2} = -M$$



An element of a beam

By integrating once, we get dy/dx along the beam, i.e. the slope of the deflection curve. By integrating twice, we get y along the beam, i.e. the shape of the deflection curve.

By differentiating the above equation with respect to x,

$EI\,(d^3y/dx^3) = -\ dM/dx = -V$ (shear)

$EI\,(d^4y/dx^4) = -\ dV/dx = w$ (load/unit length)

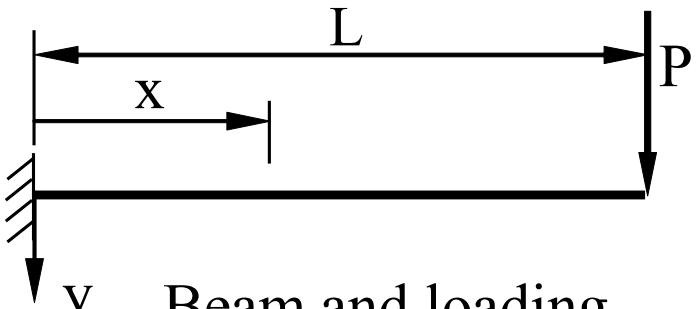
Common methods for calculating deflections

- Double integration method / Macaulay's method
- Moment-area method
- Strain energy method
- Dummy unit-load method (principle of virtual work)
- Castigliano's theorem
- Conjugate-beam method

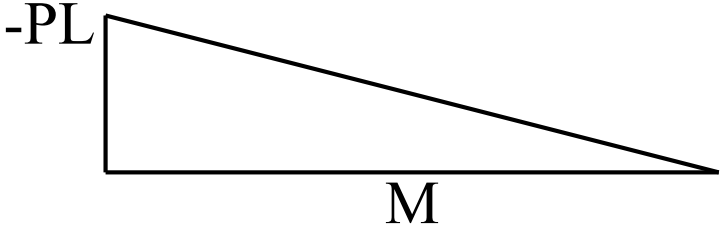
Double Integration Method



Example 1



Beam and loading



Bending moment

Cantilever under point load at tip

At x, $M_x = -P(L - x)$

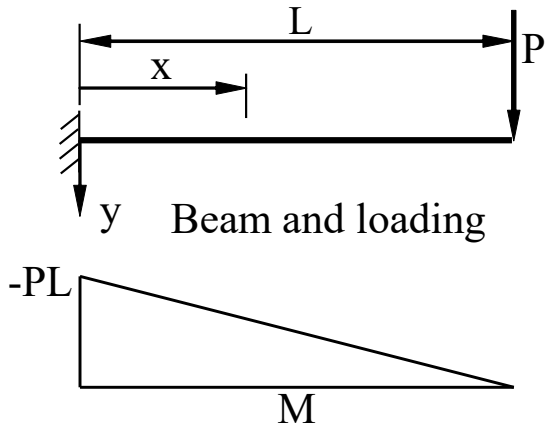
$EI\,(d^2y/dx^2) = -M_x = P(L - x)$

Integrating, $EI\,(dy/dx) = PLx - Px^2/2 + c_1$

From the boundary condition at the clamped end,

i.e. $x = 0$, $dy/dx = 0$,

$c_1 = 0$



Cantilever under point load at tip

Integrating again, $EI\,y = PLx^2/2 - Px^3/6 + c_2$

From the boundary condition at clamped end,

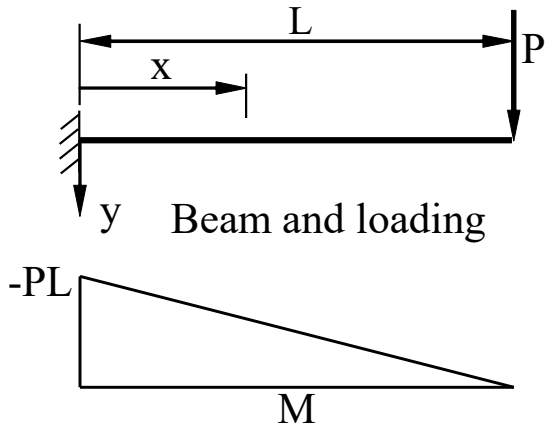
i.e. $x = 0$, $y = 0$,

$c_2 = 0$

The equations for the curve and its slope are

$y = (Px^2 / 2EI)\,(L - x/3)$

$\theta = dy/dx = (Px / EI)\,(L - x/2)$



Cantilever under point load at tip

Therefore

$$y = (Px^2 / 2EI)\,(L - x/3)$$

$$\theta = (Px / EI)\,(L - x/2)$$

At the free end $x = L$,

$$y = PL^3/3EI \qquad \text{and} \qquad \theta = PL^2 / 2EI$$

Macaulay's Method



In Macaulay's method, an expression within square brackets will be omitted *if* it is evaluated to be *negative*

- In *Double Integration Method*,
  - Separate expression required for bending moment in each segment of beam between adjacent concentrated loads or reactions
  - Each expression has its own arbitrary constant(s) of integration
- In *Macaulay's Method*
  - ONE expression for bending moment to be obtained
  - Constants of integration are the SAME for all segments of the beam

- Macaulay's bracket function

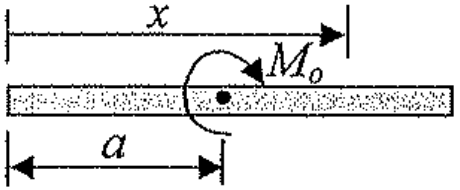
$$\left[x-a\right]^n = \begin{cases} 0 & \text{if} \quad x < a \\ \left(x-a\right)^n & \text{if} \quad x \geq a \end{cases}$$

- Integrating gives

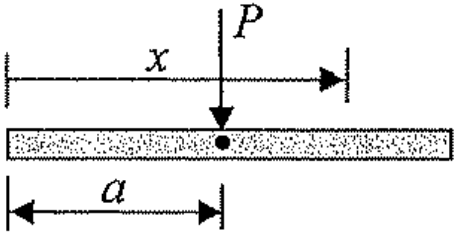
$$\int \left[x-a\right]^n dx = \frac{\left[x-a\right]^{n+1}}{n+1} + C$$

where *n* is a non-negative integer.

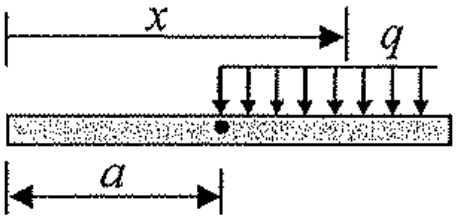
Commonly encountered loading:



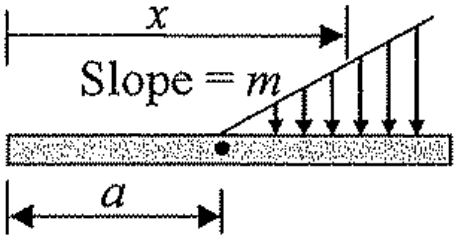
$$M = M_0 \left[x - a\right]^0$$



$$M = -P\left[x - a\right]$$

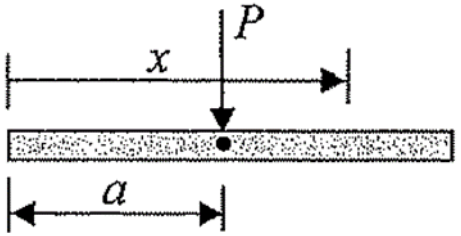


$$M = -\frac{q}{2}\left[x - a\right]^2$$

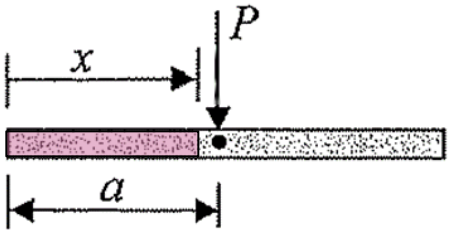


$$M = -\frac{m}{6}\left[x - a\right]^3$$

- Commonly encountered loading:

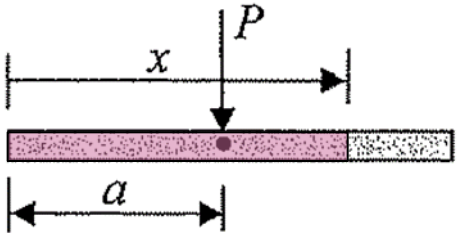


$$M = -P\left[x - a\right]$$



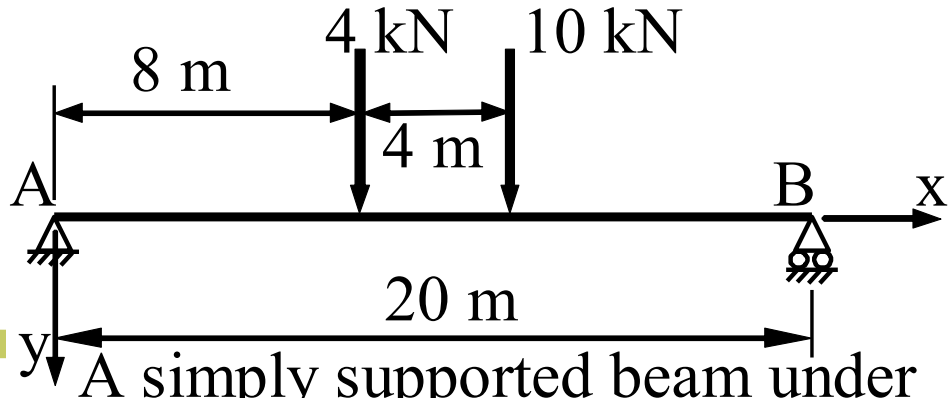
Two different scenarios:

If $x < a$, $M = 0$



If $x \geq a$, $M = -P\,(x - a)$

Example 2

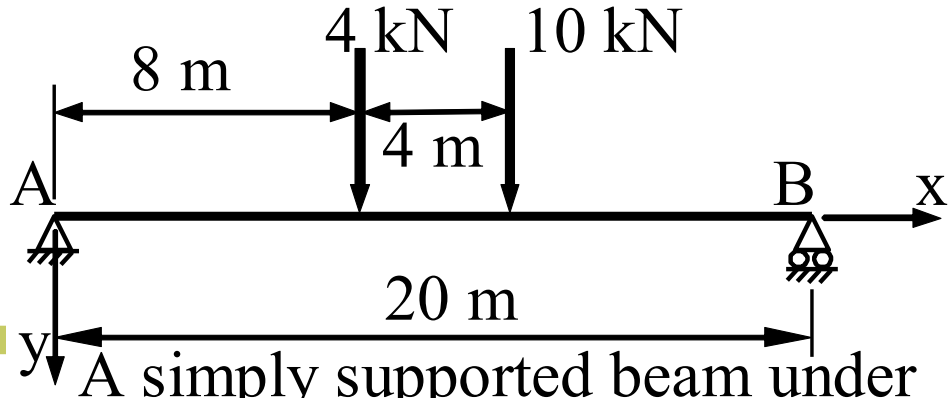


A simply supported beam under two point loads

A simply supported beam of span 20 m carries two point loads of 4 kN at 8 m and 10 kN at 12 m from the left end.

Calculate the deflection under each point load.

Take E = 200000 N/mm^2 and I = 10^9 mm^4.



A simply supported beam under two point loads

Reaction at A $= (4 \times 12 + 10 \times 8) / 20 = 6.4$ kN

$-EI\,(d^2y/dx^2) = M_x$

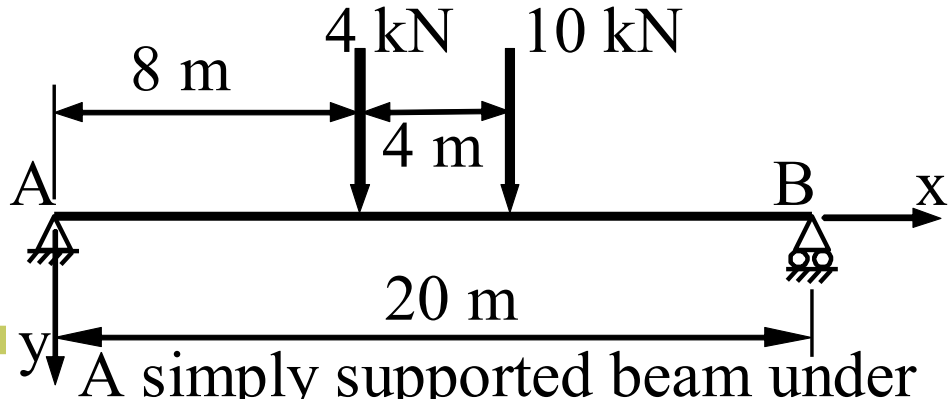
$= 6.4\,x - 4\,[x - 8] - 10\,[x - 12]$

Integrating

$-EI\,(dy/dx) = 3.2\,x^2 - 2\,[x - 8]^2 - 5\,[x - 12]^2 + A$

Integrating again

$-EI\,y = (3.2/3)\,x^3 - (2/3)\,[x - 8]^3 - (5/3)\,[x - 12]^3 + Ax + B$



A simply supported beam under two point loads

When x = 0 m, y = 0 m,

$\therefore$ B = 0 kNm^3

When x = 20 m,

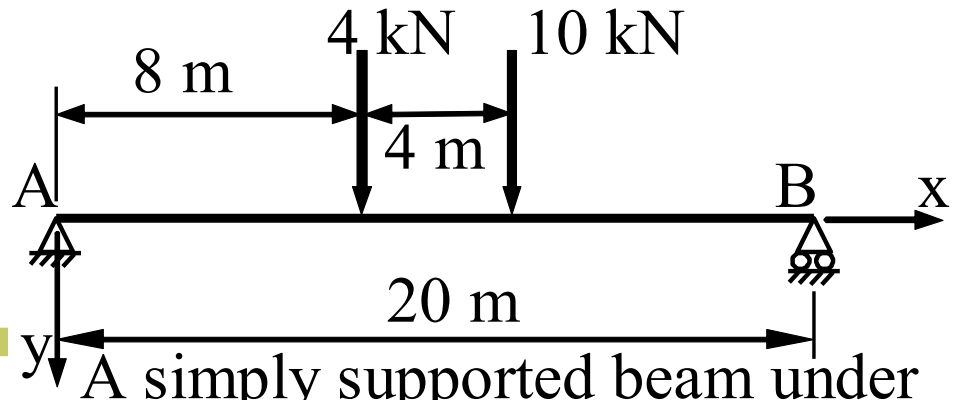
$y = (3.2/3)\,(20)^3 - (2/3)\,(12)^3 - (5/3)\,(8)^3 + A\,(20)$

$= 0$

giving

$A = -(3.2/3)\,(20)^2 + (2/3)\,(12)^3 / 20 + (5/3)\,(8)^3 /20$

$= -326.4$ kNm^2



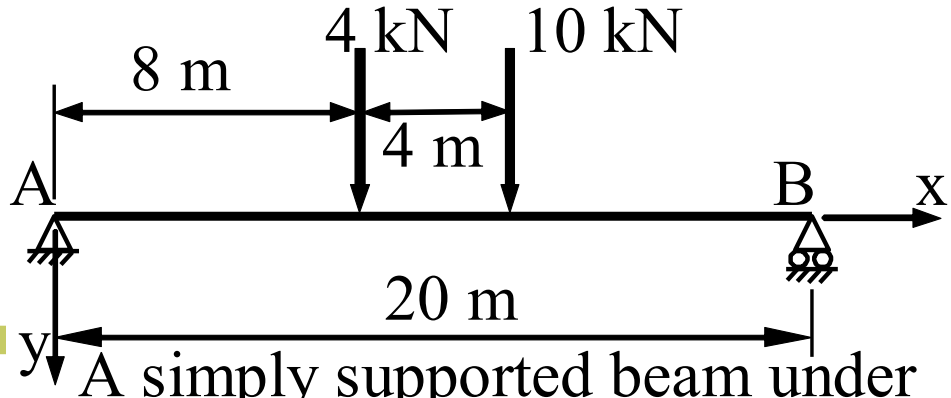
A simply supported beam under two point loads

Deflection under the 4 kN point load (x = 8 m):

$$-EI\,y = (3.2/3)\,(8)^3 + (-326.4)\,(8) = -2065\ \text{kNm}^3$$

$$y = (2065 \times 10^3 \times 10^9) / (200000 \times 10^9)$$

$$= 10.3\ \text{mm (downwards)}$$



A simply supported beam under two point loads

Deflection under the 10 kN point load (x = 12 m):

$$-EI\,y = (3.2/3)\,(12)^3 - (2/3)\,(4)^3 + (-326.4)\,(12)$$

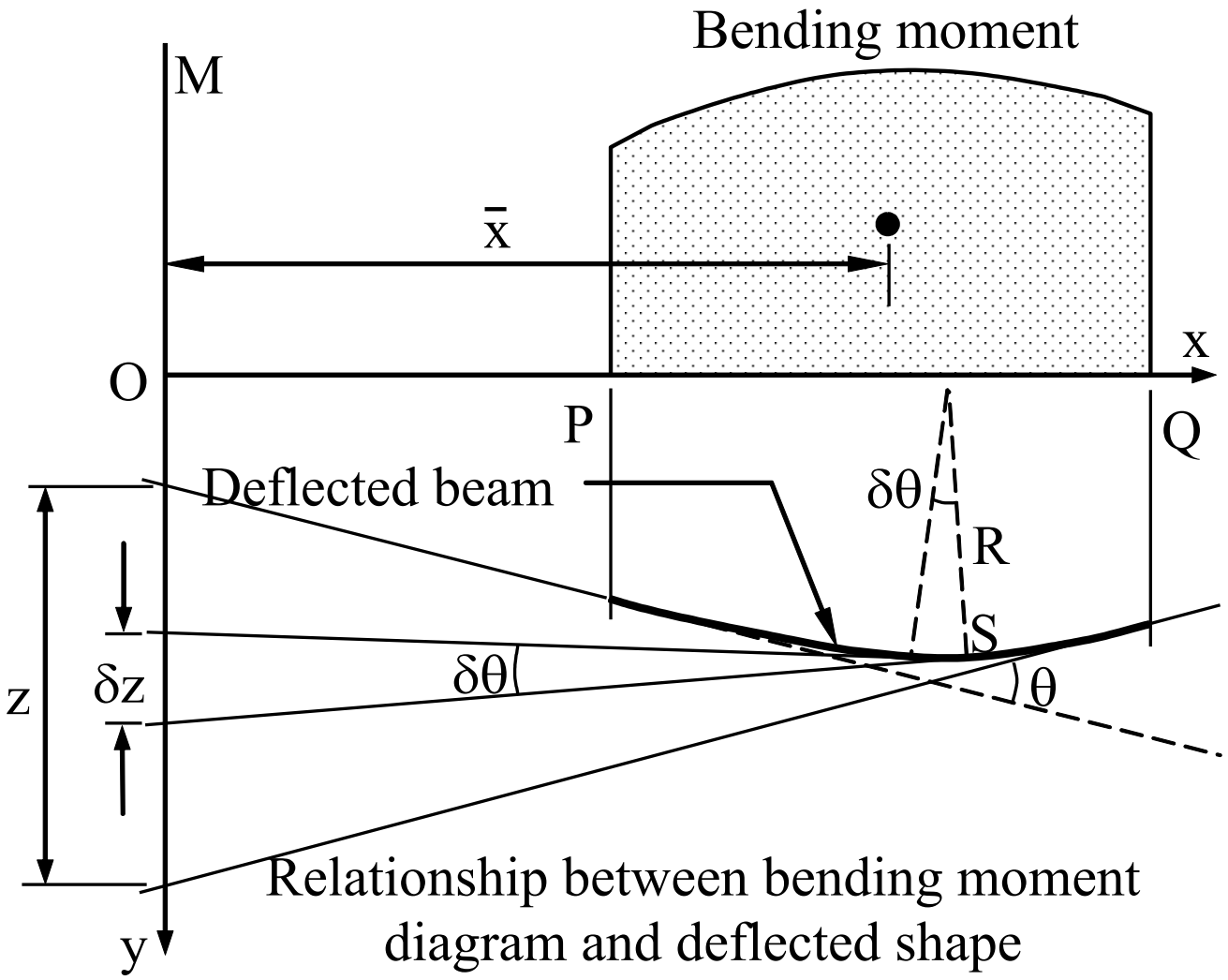
$$= -2116\ \text{kNm}^3$$

$$y = (2116 \times 10^3 \times 10^9) / (200000 \times 10^9)$$

$$= 10.6\ \text{mm}\ \text{(downwards)}$$

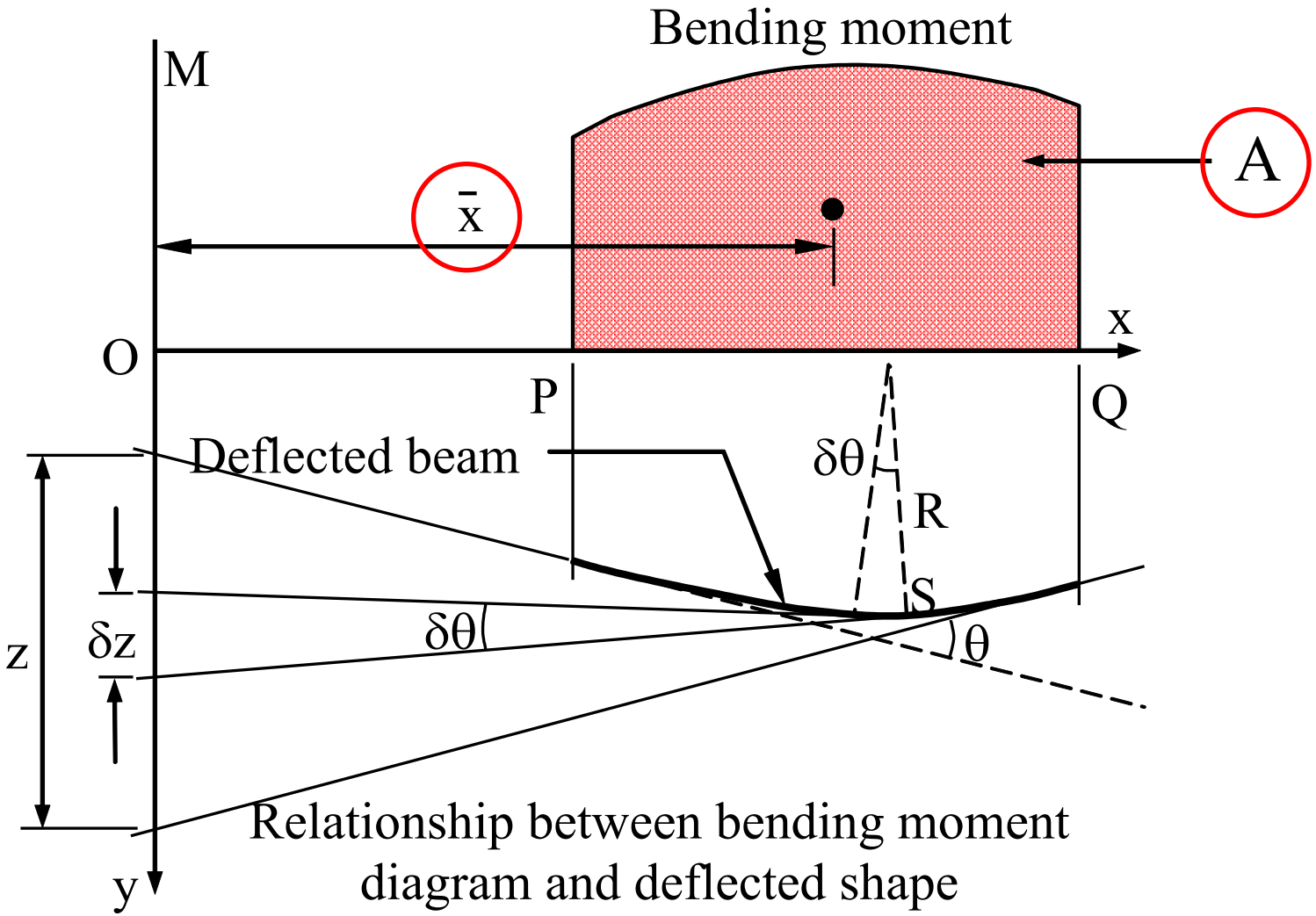
Moment-area Method





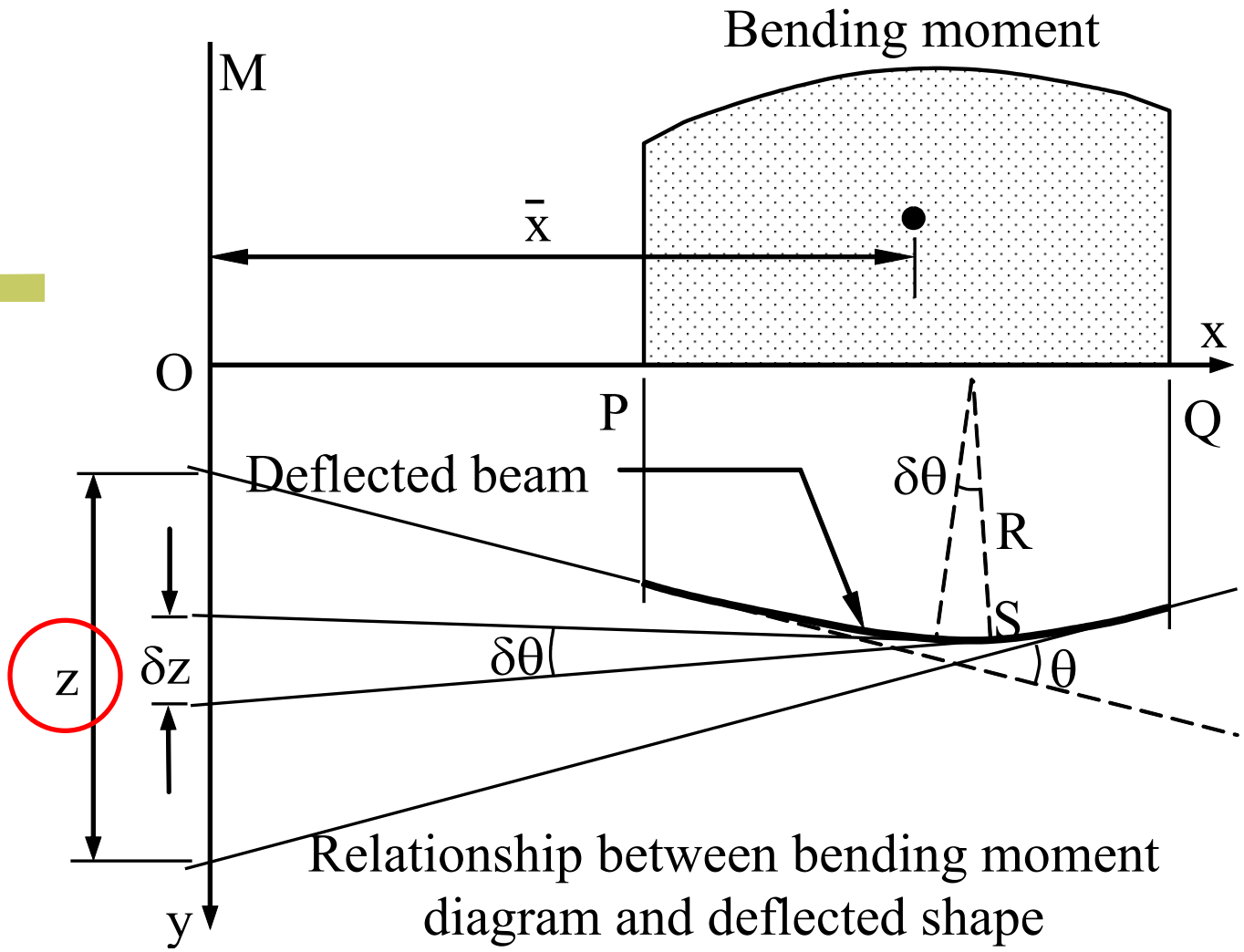
Relationship between bending moment diagram and deflected shape

The above shows the bending moment diagram and the shape of the deflected beam between two arbitrary points P and Q.



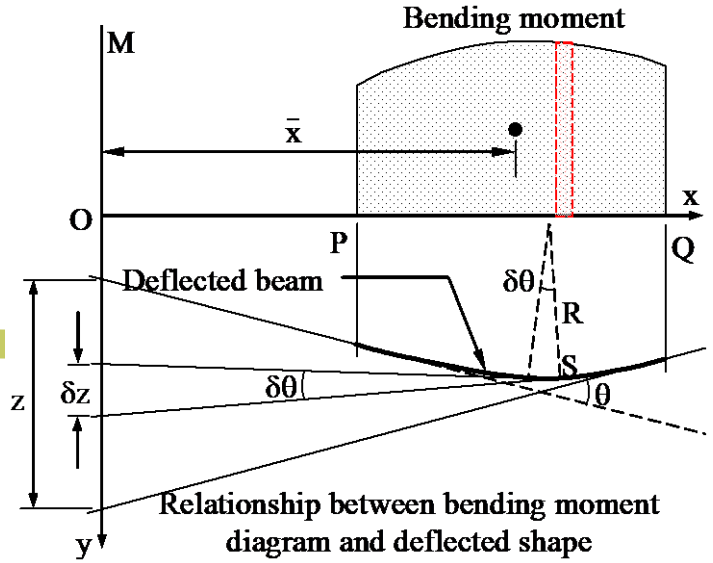
Relationship between bending moment diagram and deflected shape

- The area of the bending moment diagram between P and Q is A, and its centroid is at a distance $\bar{x}$ from line Oy.



Relationship between bending moment diagram and deflected shape

- The tangents at P and Q to the elastic line cut off an intercept z on Oy.



$$\frac{d^2y}{dx^2} = -\frac{M}{EI}$$

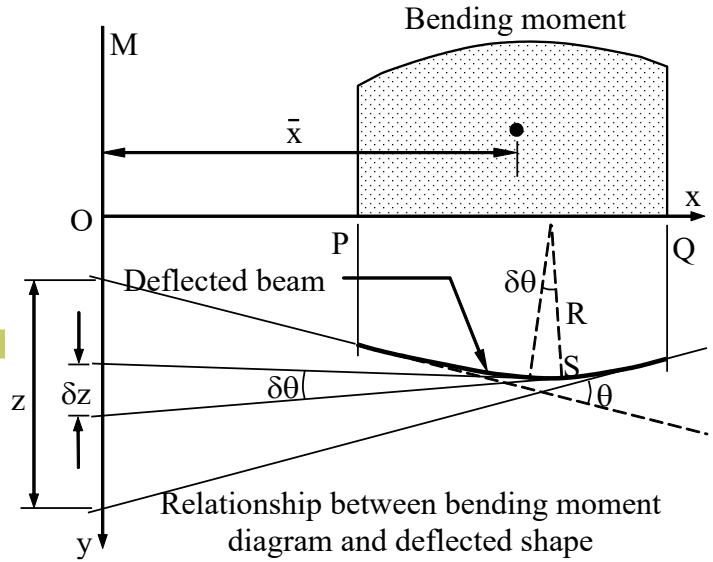
Integrating between P and Q,

$$\left[\frac{dy}{dx}\right]_P^Q = -\int \frac{Mdx}{EI}$$

If EI is constant,

$$\theta = \left(\frac{dy}{dx}\right)_P - \left(\frac{dy}{dx}\right)_Q = \frac{A}{EI}$$

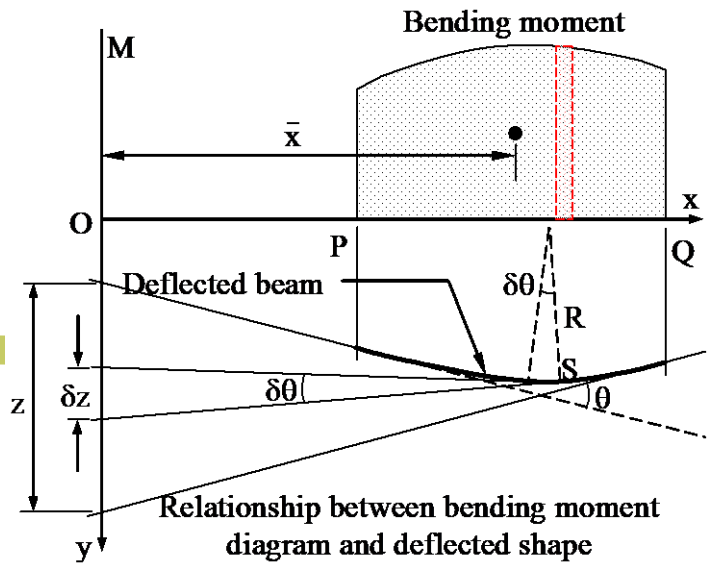
Therefore the angle made by the tangents to two points on a beam is equal to the net area of the bending moment diagram between those points divided by EI.



At an arbitrary point S between P and Q,

- radius of curvature of the deflected beam is R
- angle between the tangents at the ends of a short length δx is $\delta\theta$, where $\delta x = R \cdot \delta\theta$
- intercept of these tangents on Oy is δz, and because the slope is everywhere small

$$\delta z = x \cdot \delta\theta = x \cdot \delta x / R = Mx \cdot \delta x / EI$$



Integrating between P and Q,

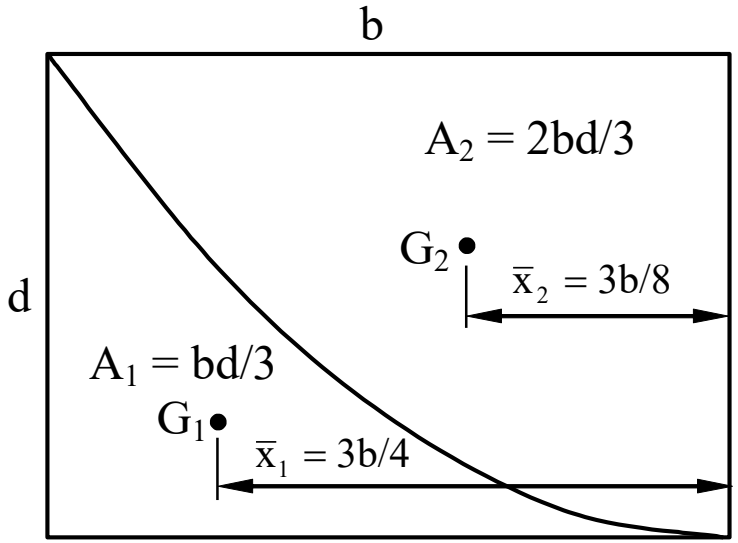
$$z = \int \frac{Mx\,dx}{EI} = \frac{A\bar{x}}{EI}$$

if EI is constant

Therefore the intercept on a given line between the tangents to the beam at any points P and Q is equal to the net moment about that line of the bending moment diagram between P and Q divided by EI.

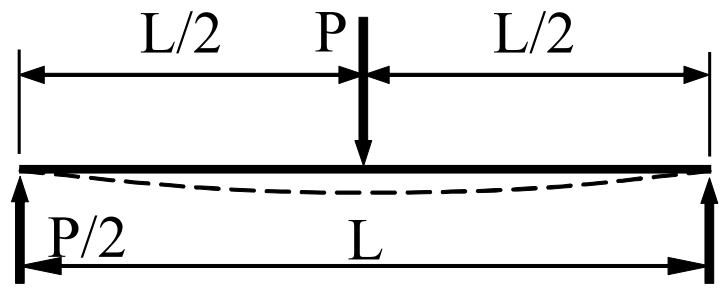
Moment-area Method

- It is convenient to break down the bending moment diagram into a number of simple figures so that the moment is obtained from $\sum A\bar{x}$
- Account should be taken of positive and negative areas.

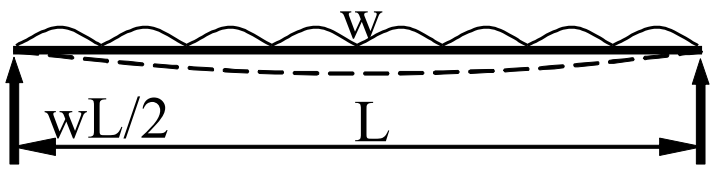


Properties of common shapes

Example 3



(a) A simply supported beam under a central point load



(b) A simply supported beam under uniformly distributed load

Obtain expressions for the maximum slope and deflection of a simply supported beam of span L

a) with a concentrated load P at mid-span

b) with a uniformly distributed load w over the whole span.

Take the flexural rigidity EI as constant

Concentrated load P at mid-span

$A = (1/2)\,(PL/4)\,(L/2) = PL^2/16$

Slope at support $= A/EI = PL^2/16EI$

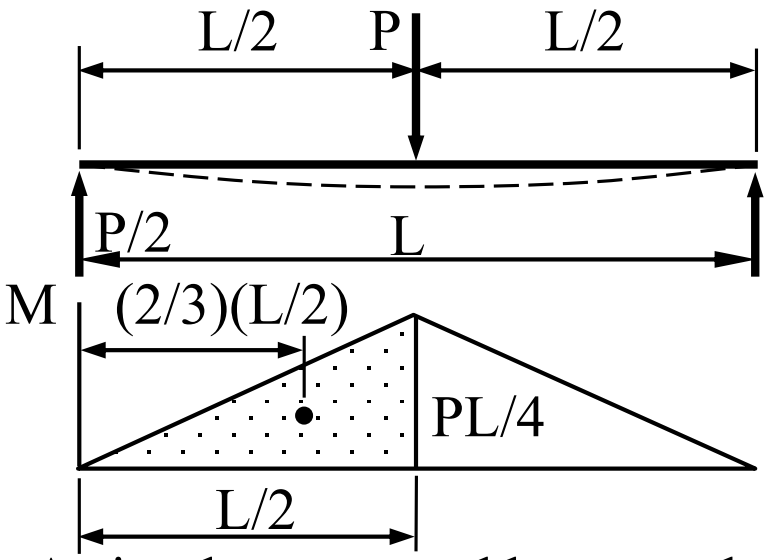
Deflection at mid-span

= deflection of support relative to tangent to centre

$= A\bar{x}/EI$

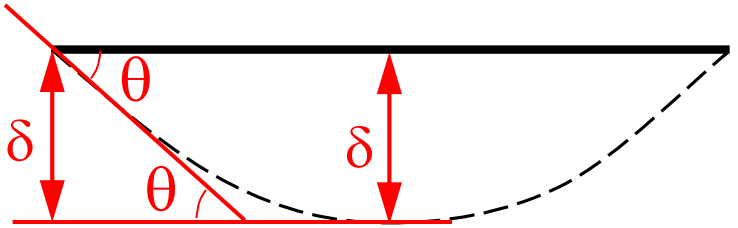
$= (PL^2/16)\,(L/3) / EI$

$= PL^3/48EI$



A simply supported beam under a central point load

Deflection exaggerated



Uniformly distributed load w over the whole span

$A = (2/3)(wL^2/8)(L/2) = wL^3/24$

Slope at support $= A/EI = wL^3/24EI$

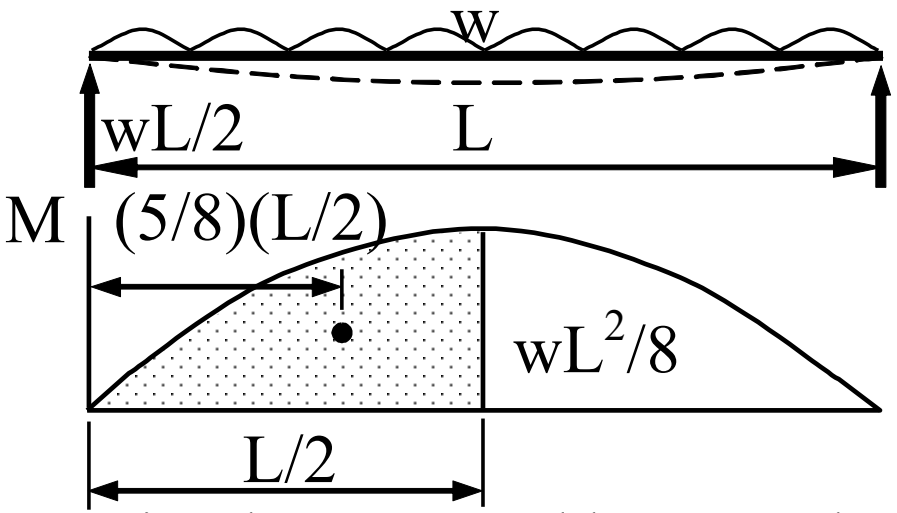
Deflection at mid-span

= deflection of support relative to tangent to centre

$= A\bar{x}/EI$

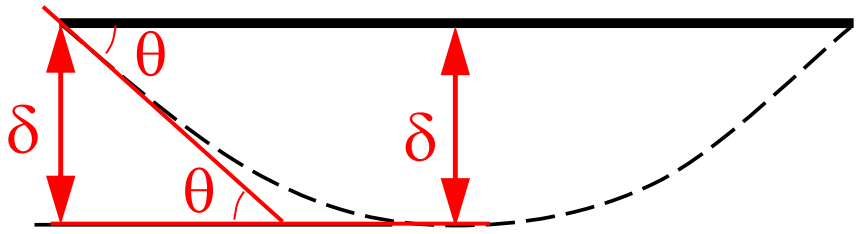
$= (wL^3/24)(5L/16) / EI$

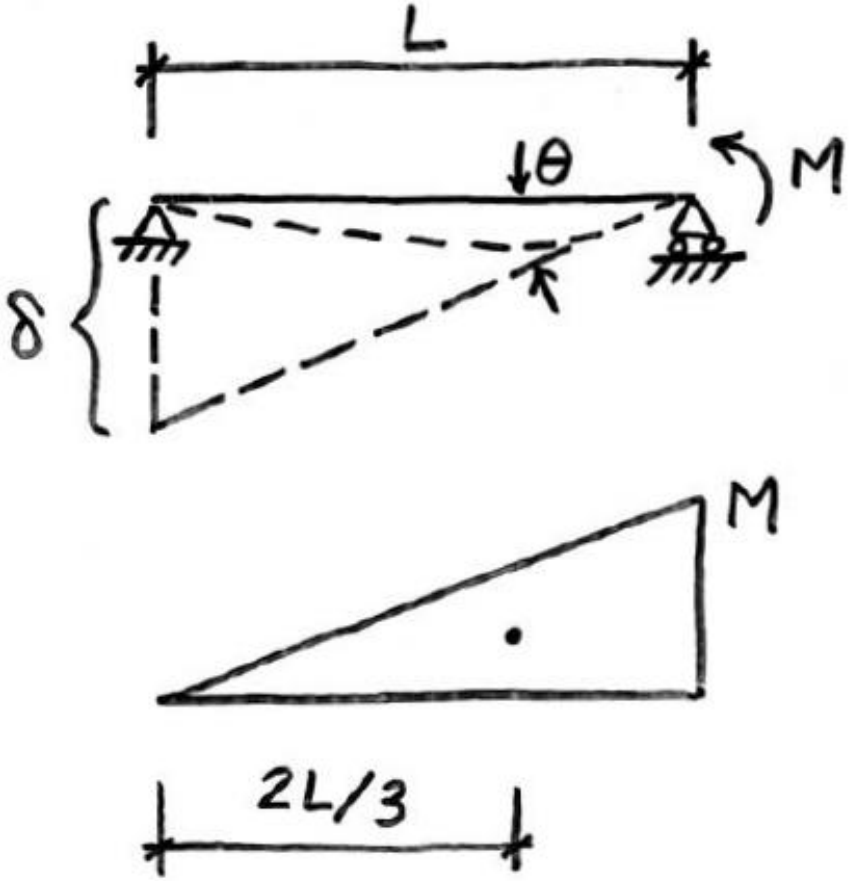
$= 5wL^4/384EI$



A simply supported beam under uniformly distributed load

Deflection exaggerated





$$\delta = \frac{ML}{2EI} \cdot \frac{2L}{3} = \frac{ML^2}{3EI}$$

$$\theta = \frac{\delta}{L} = \frac{ML}{3EI}$$

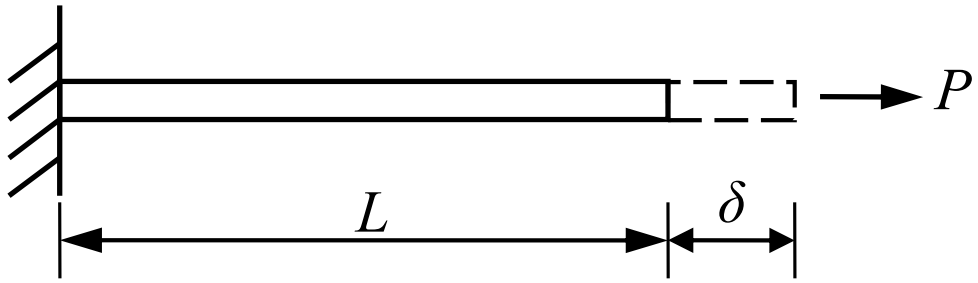
Strain Energy Method

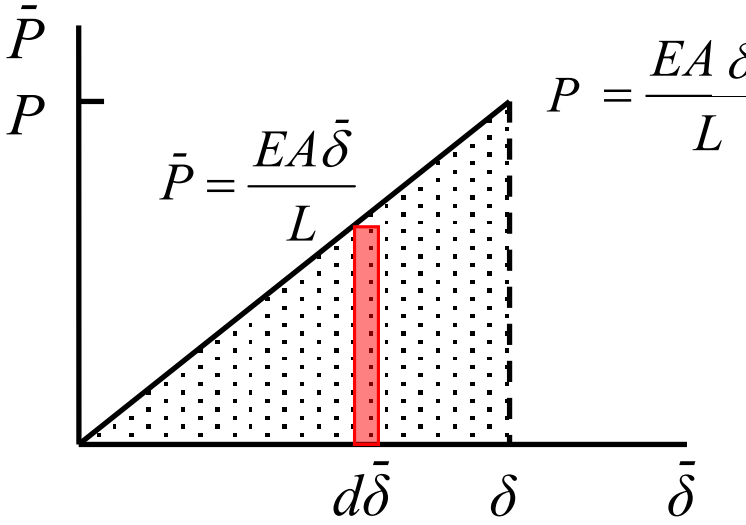


Strain Energy Method

- When an elastic body is deformed by external forces, these forces move through small distances and hence do some work
- The energy of the elastic system has been increased by the work done. This work is
  - recoverable
  - stored in the elastic body in the form of elastic potential energy or *strain energy*

The energy concept can be used to find the deflections of a structure





$$P = \frac{EA\,\delta}{L}$$

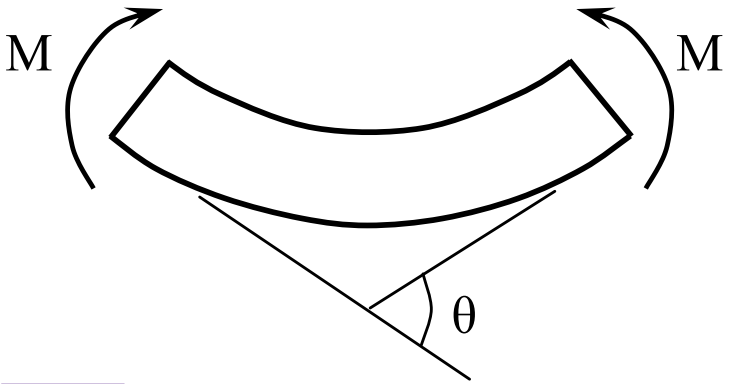
$$U = \int_0^{\delta} \frac{EA\bar{\delta}}{L} d\bar{\delta}$$

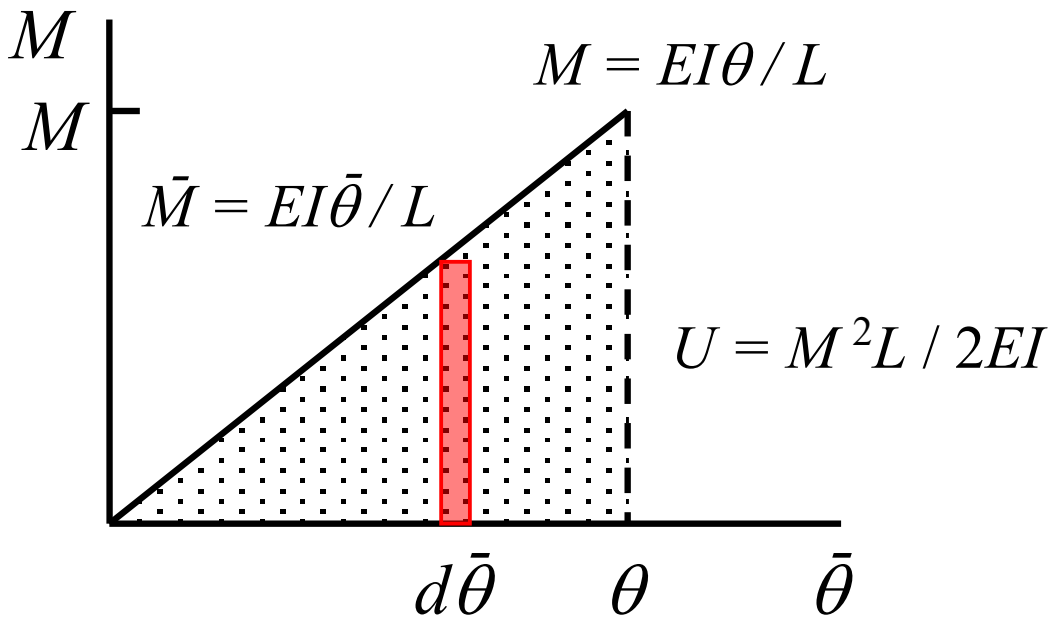
$$= \left[\frac{EA\bar{\delta}^2}{2L} \right]_0^{\delta}$$

$$= \frac{P^2 L}{2EA}$$

Consider a prismatic bar of length L and cross-sectional area A under a gradually applied axial load P

- It will elongate by $\delta = PL/EA$
- The work done by P is $P(\delta/2) = P^2L / 2EA$, which is the strain energy stored in the bar

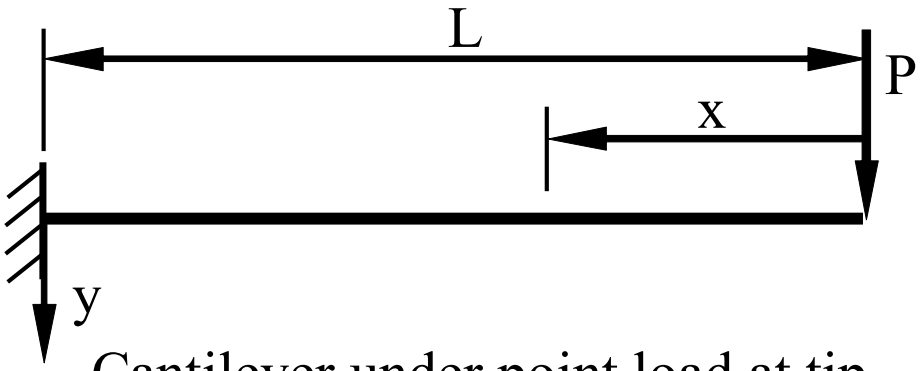




Consider a prismatic beam of length L and second moment of area I under the action of gradually applied pure bending moment M

- It will be bent such that the two ends form an angle of θ given by $\theta = ML / EI$
- The work done by M is $M(\theta/2) = M^2L/2EI$, which is the strain energy in the beam

Example 4



Cantilever under point load at tip

For a small segment dx at x from the free end,

$M_x = Px$. The strain energy is

$$\Delta U = \frac{(Px)^2 dx}{2EI}$$

$$U = \int_0^L \frac{P^2 x^2}{2EI} dx = \frac{P^2}{2EI}\left[\frac{x^3}{3}\right]_0^L = \frac{P^2 L^3}{6EI}$$

The work done by the load P in deflecting a distance δ is $P\delta/2$, which must be equal to the energy stored $P^2L^3 / 6EI$. Therefore $\delta = PL^3 / 3EI$

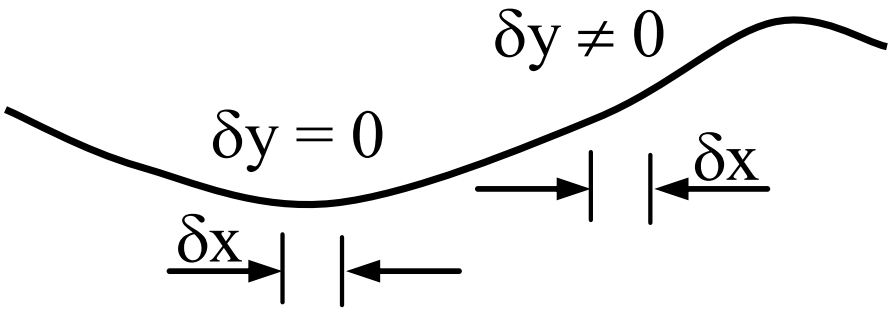
Dummy Unit-load Method

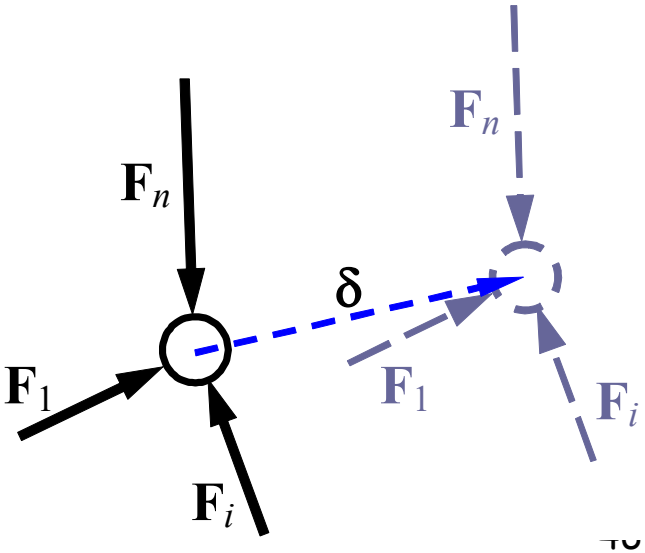


Dummy unit-load method

The *principle of virtual work* states that when a complete system of forces, which is in equilibrium, undergoes a pattern of geometrically consistent displacements such that the magnitudes and directions of all of the forces are unchanged, the total virtual work done is zero

Virtual work = $\Sigma F \delta = 0$



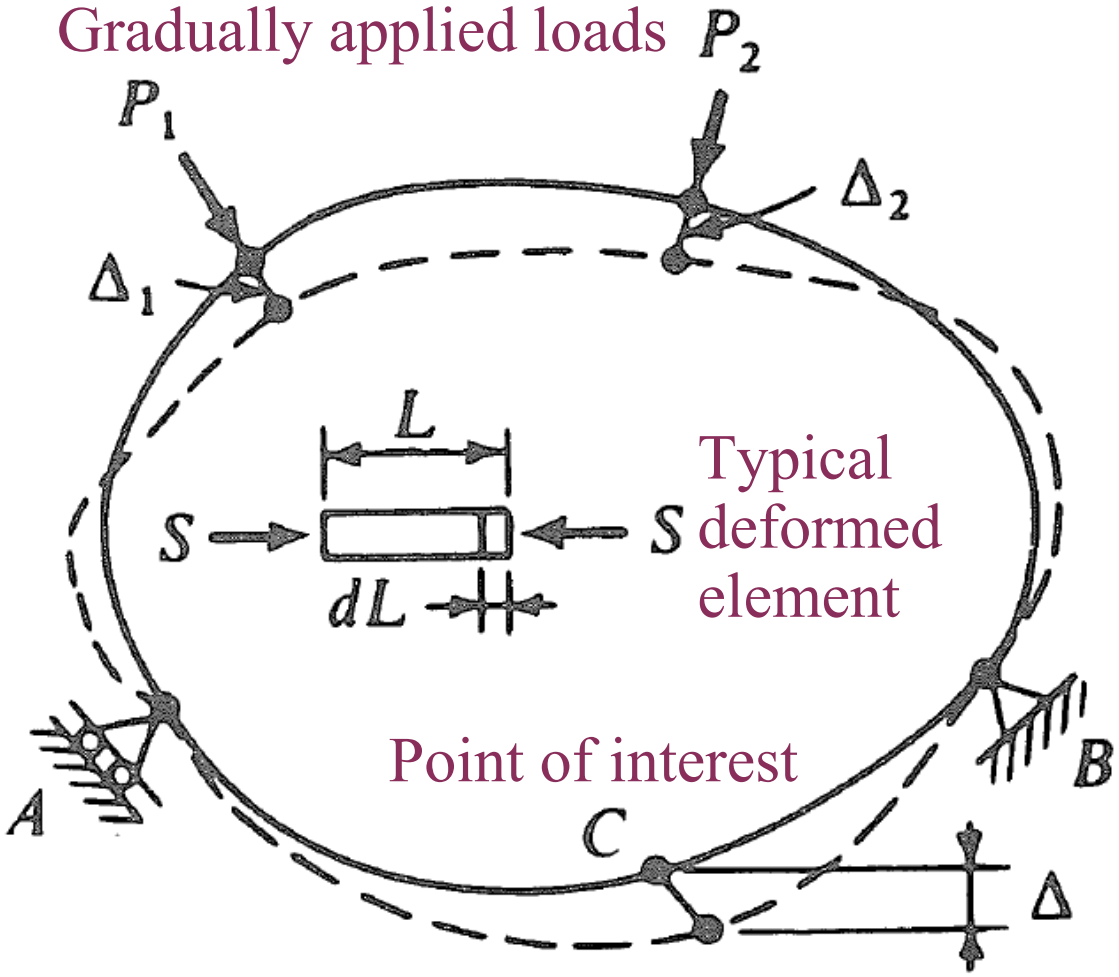


Dummy unit-load method

- Virtual work is established in one of the following ways
  1. By considering the movements of the actual force system through some fictitious pattern of displacements (*virtual displacements*)
  2. By considering the movements of some fictitious system of forces (*virtual forces*) through the actual displacement pattern

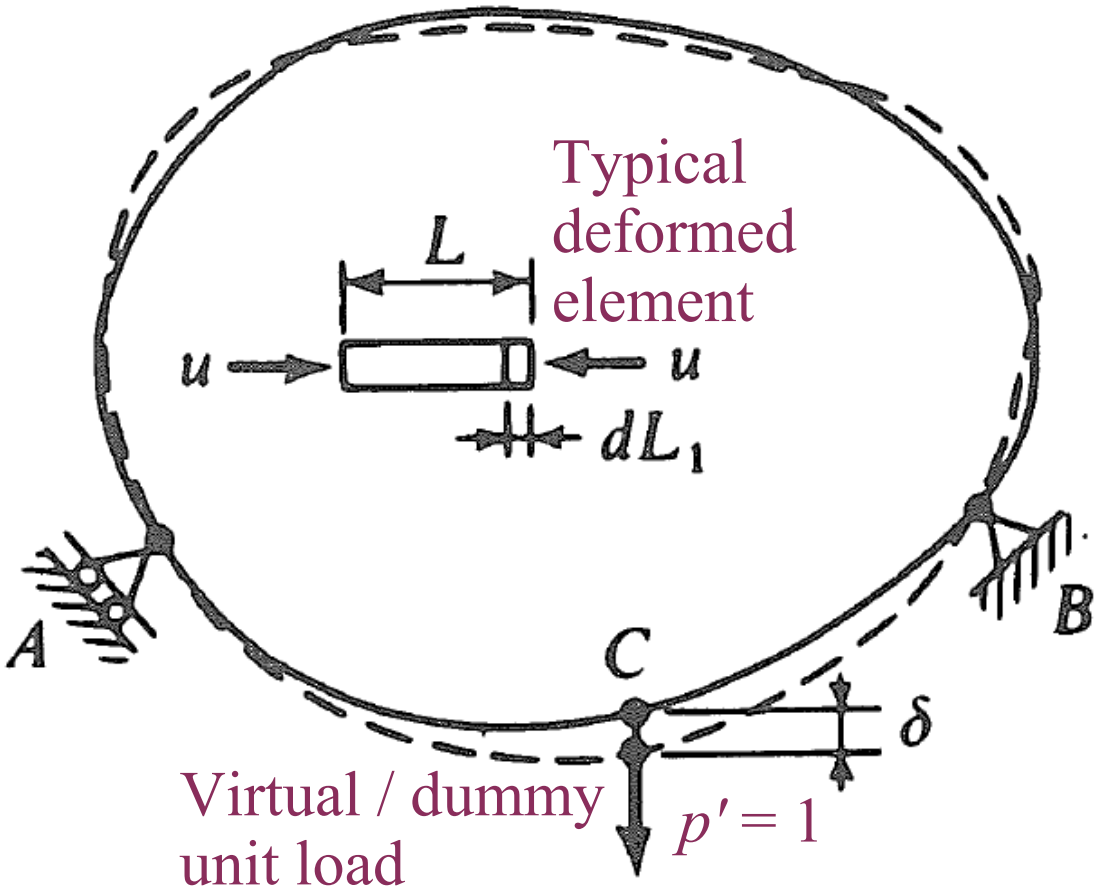
One component of the deflection or rotation at a point can be found in each application of the unit load method

A deformed elastic structure under gradually applied loads



$$\tfrac{1}{2} P_1\Delta_1 + \tfrac{1}{2} P_2\Delta_2 = \tfrac{1}{2} \Sigma S\, dL \qquad (1)$$

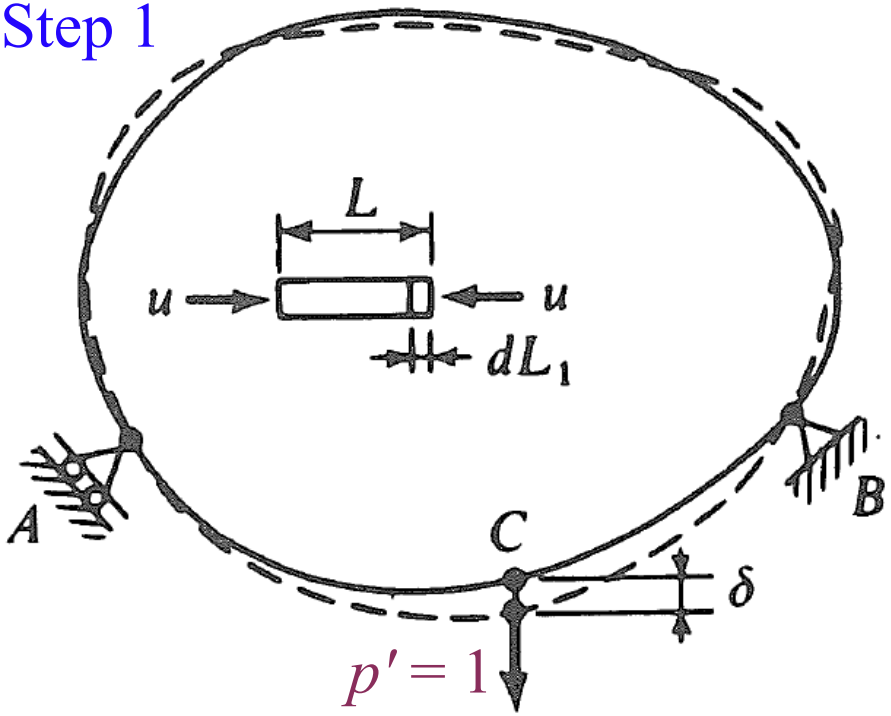
A deformed elastic structure under virtual unit load



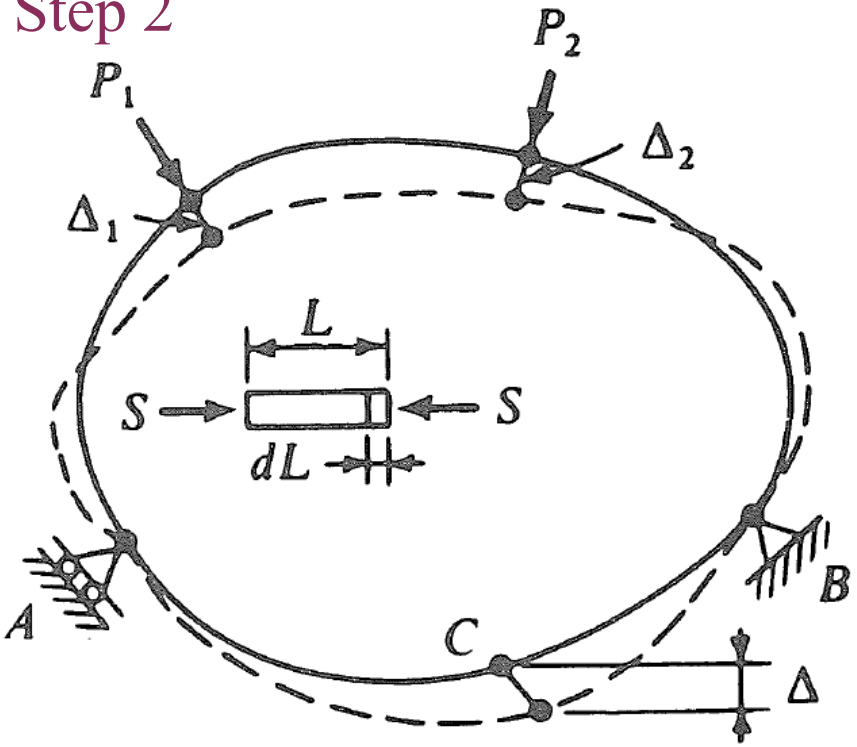
$$\tfrac{1}{2}\, p'\, \delta = \tfrac{1}{2}\, \Sigma\, u\, dL_1 \qquad (2)$$

Imagine this …

Step 1



Step 2



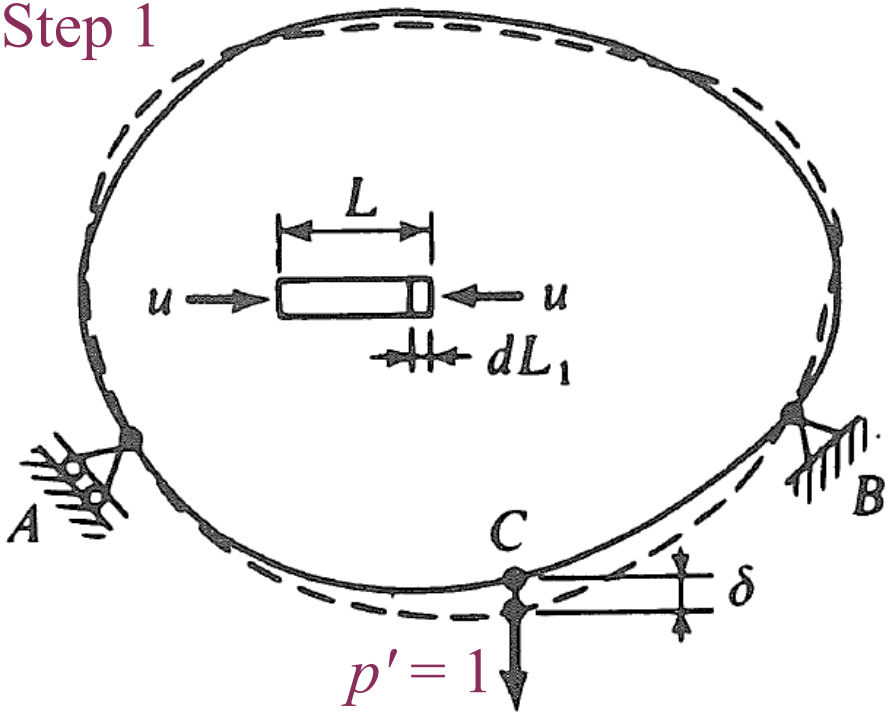
Equating total work done and total strain energy,

$$\tfrac{1}{2}\, p'\delta + \tfrac{1}{2}\, P_1\Delta_1 + \tfrac{1}{2}\, P_2\Delta_2 + p'\Delta$$

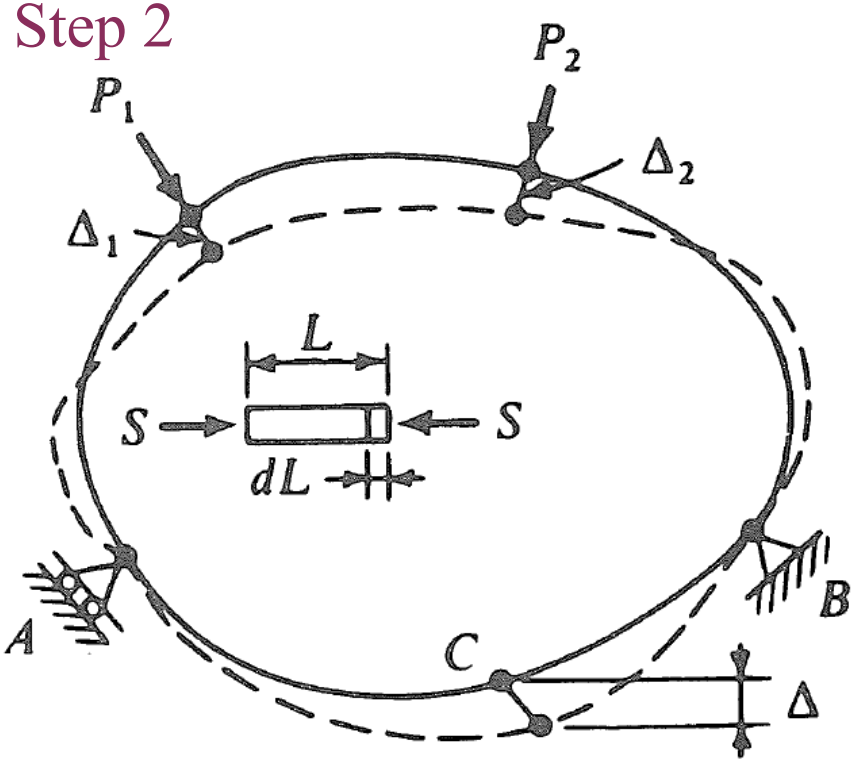
$$= \tfrac{1}{2}\, \Sigma\, u\, dL_1 + \tfrac{1}{2}\, \Sigma\, S\, dL + \Sigma\, u\, dL \qquad (3)$$

Imagine this …

Step 1



Step 2



Subtracting the sum of (1) and (2) from (3) gives

$p' \Delta = \Sigma\, u\, dL$ or $1 \cdot \Delta = \Sigma\, u \cdot dL$

virtual — actual

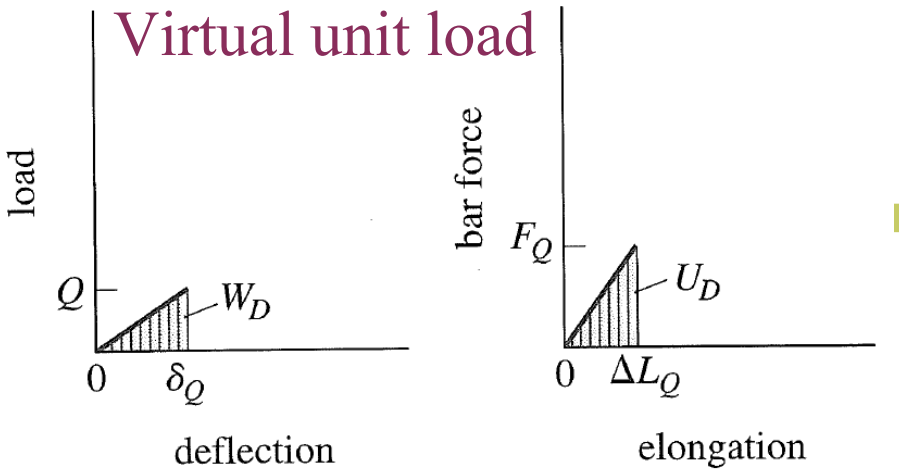
For calculation of rotation, apply unit moment

$1 \cdot \theta = \Sigma\, u \cdot dL$

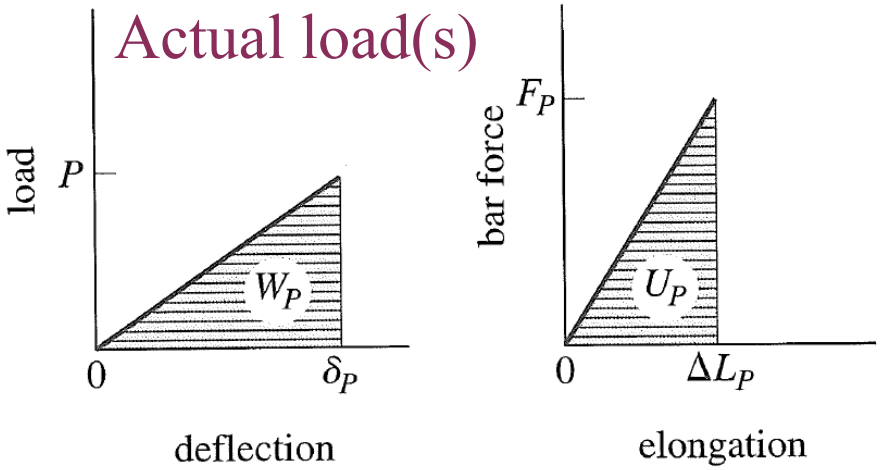


Analogy of a bar

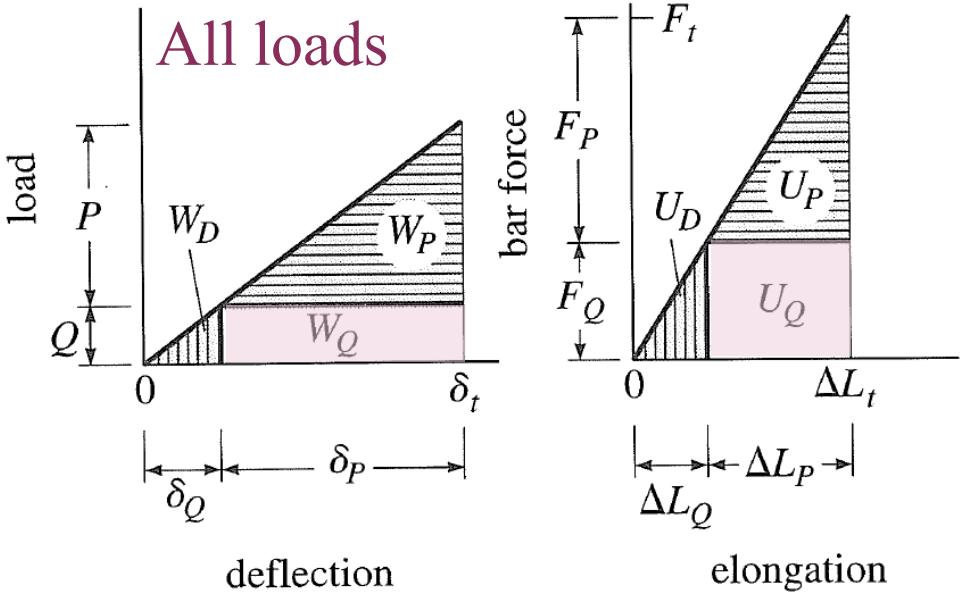
Virtual unit load



Actual load(s)



All loads

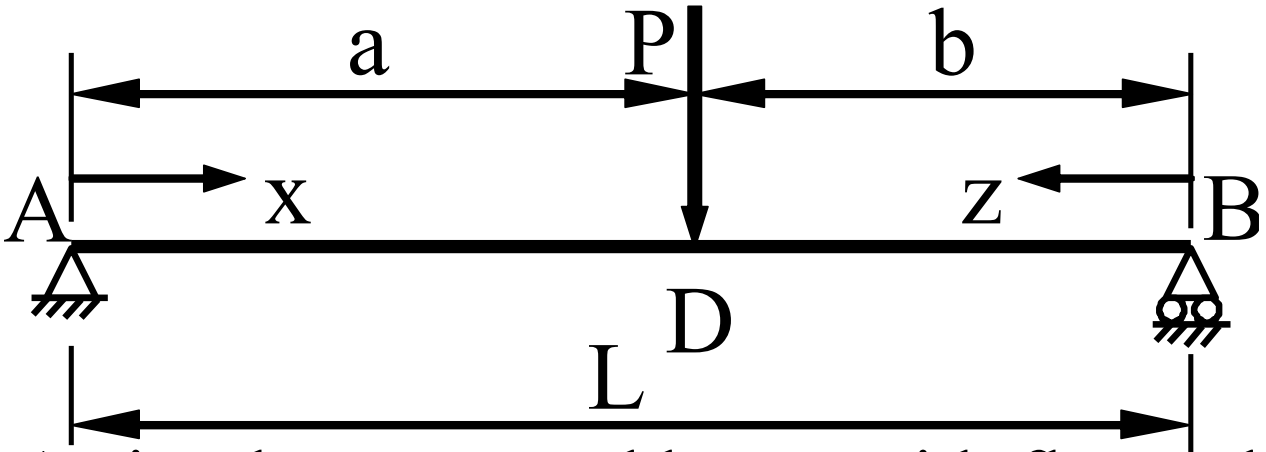


Virtual work = Virtual internal energy

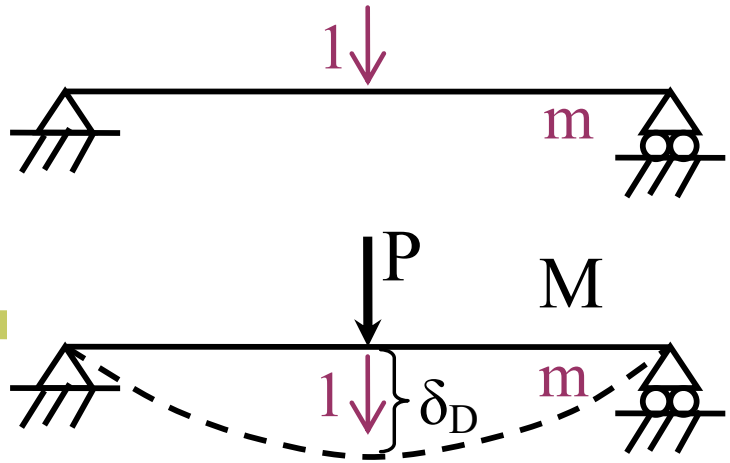
$1 \cdot \Delta = \Sigma\, u \cdot dL$

Example 5

- Find the deflection δ_D at D of the simply supported beam AB



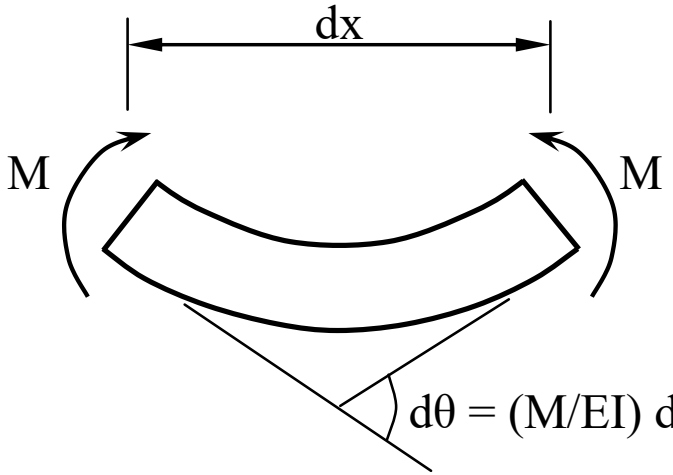
A simply supported beam with flexural rigidity EI under a point load P at D

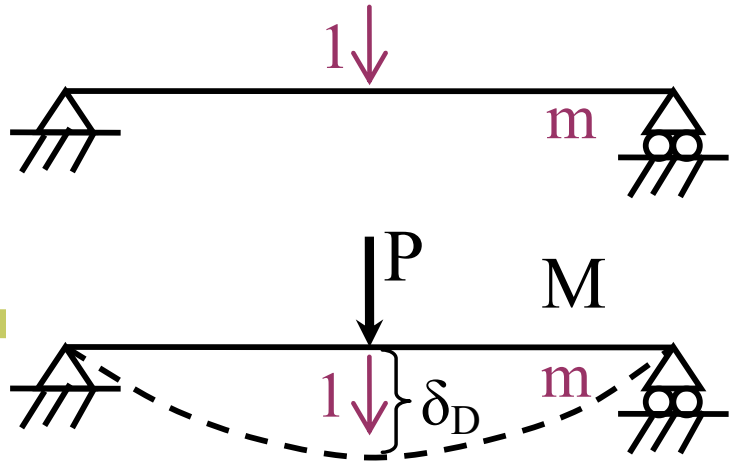


1. Place a dummy (virtual) unit-load at D, and consequently bending moment *m* is induced throughout the structure.

   Note that the displacement caused by this unit load is insignificant.

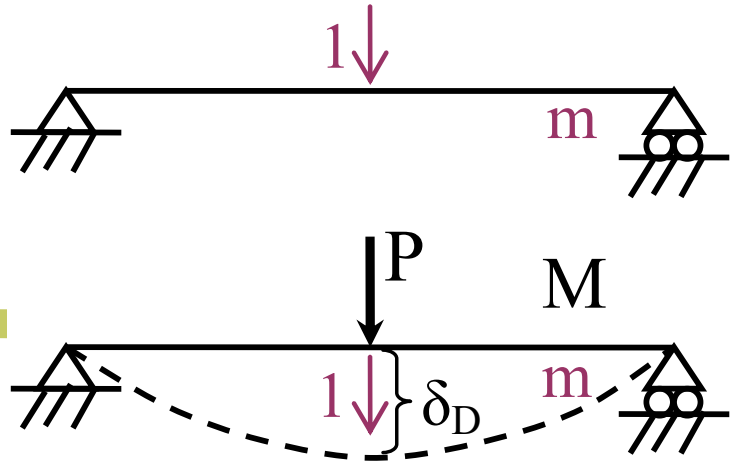
2. Impose the applied loading system *P* (actual), and let *M* be the bending moment in the structure





3. Due to the actual loading system P, the deflection at D is δ_D and the relative rotation of any two sections dx apart is $d\theta = (M/EI)\,dx$
4. Considering the bending energy only, the virtual work stored due to moment m through an angle $d\theta = dU = m\,(M/EI)\,dx$
5. The total virtual internal energy is

$$U \approx \int (Mm/EI)\,dx$$

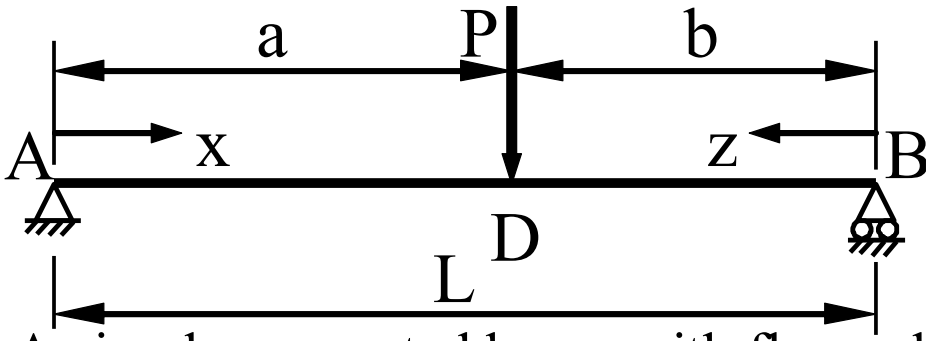


The external work done (virtual) by the dummy load is: $1 \times \delta_D$

Since external work done = internal energy stored (all virtual),

$$1 \times \delta_D = \int (Mm/EI)\, dx \quad \text{or} \quad \delta_D = \int (Mm/EI)\, dx$$

Member	M	m	Limits
AD	(Pb/L)x	(b/L)x	0 - a
BD	(Pa/L)z	(a/L)z	0 - b



A simply supported beam with flexural rigidity EI under a point load P at D

$$\delta_D = \int (Mm/EI)dx$$

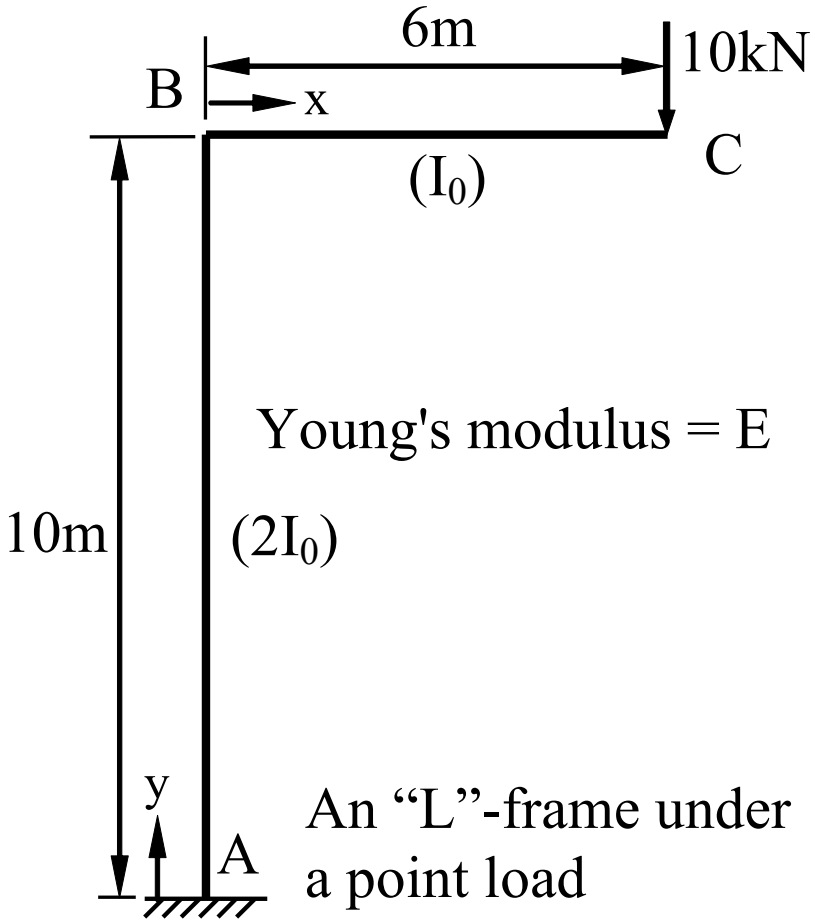
$$= \int_0^a (1/EI)\,(Pb/L)\,x\,(b/L)\,x\,dx + \int_0^b (1/EI)\,(Pa/L)\,z\,(a/L)\,z\,dz$$

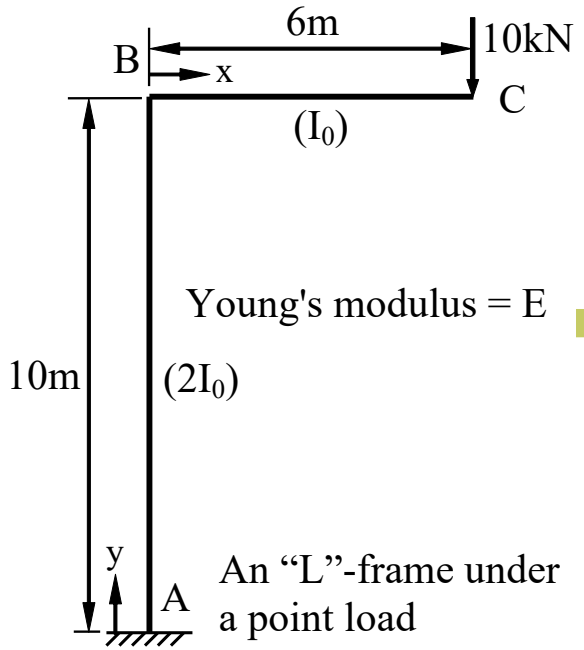
$$= \left(\frac{1}{EI}\right)P\left(\frac{b^2}{L^2}\right)\left(\frac{a^3}{3}\right) + \left(\frac{1}{EI}\right)P\left(\frac{a^2}{L^2}\right)\left(\frac{b^3}{3}\right)$$

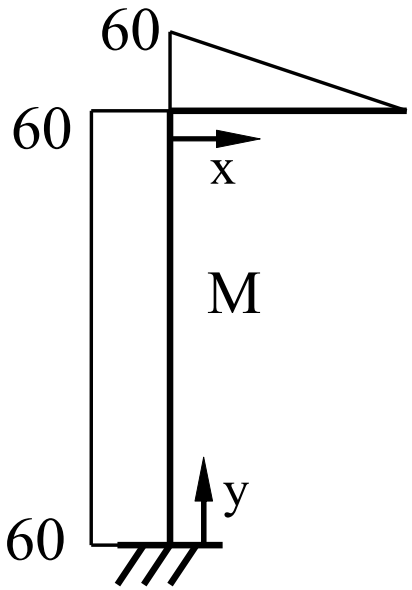
$$= \frac{P(a^2b^2)(a+b)}{3EIL^2} = \frac{Pa^2b^2}{3EIL}$$

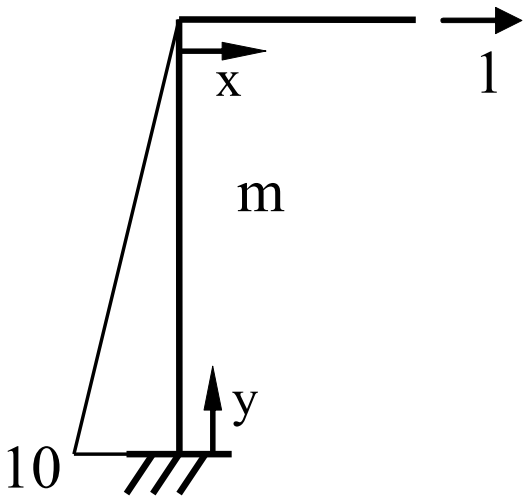
Example 6

- Find the horizontal deflection at the free end of a cantilever bent ABC









- Apply a dummy unit-load horizontally at C

$$\text{AB: } \int (Mm/EI)\,dy = \int_0^{10} \frac{(60)(10-y)}{2EI_0}\,dy = \frac{1500}{EI_0}$$

$$\text{BC: } \int (Mm/EI)\,dx = \int_0^{6} \frac{10(6-x)(0)}{EI_0}\,dx = 0$$

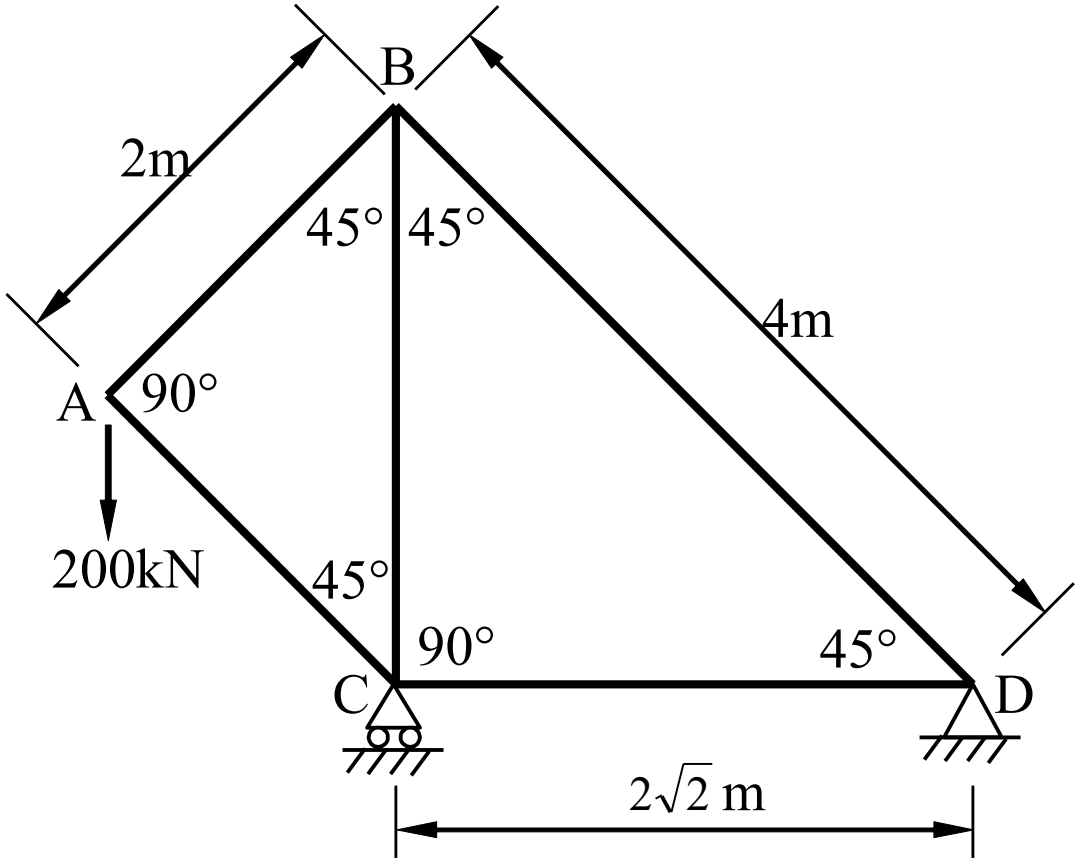
$$\delta_{HC} = \frac{1500}{EI_0} \quad \text{(in metres)}$$

Deflection of Trusses

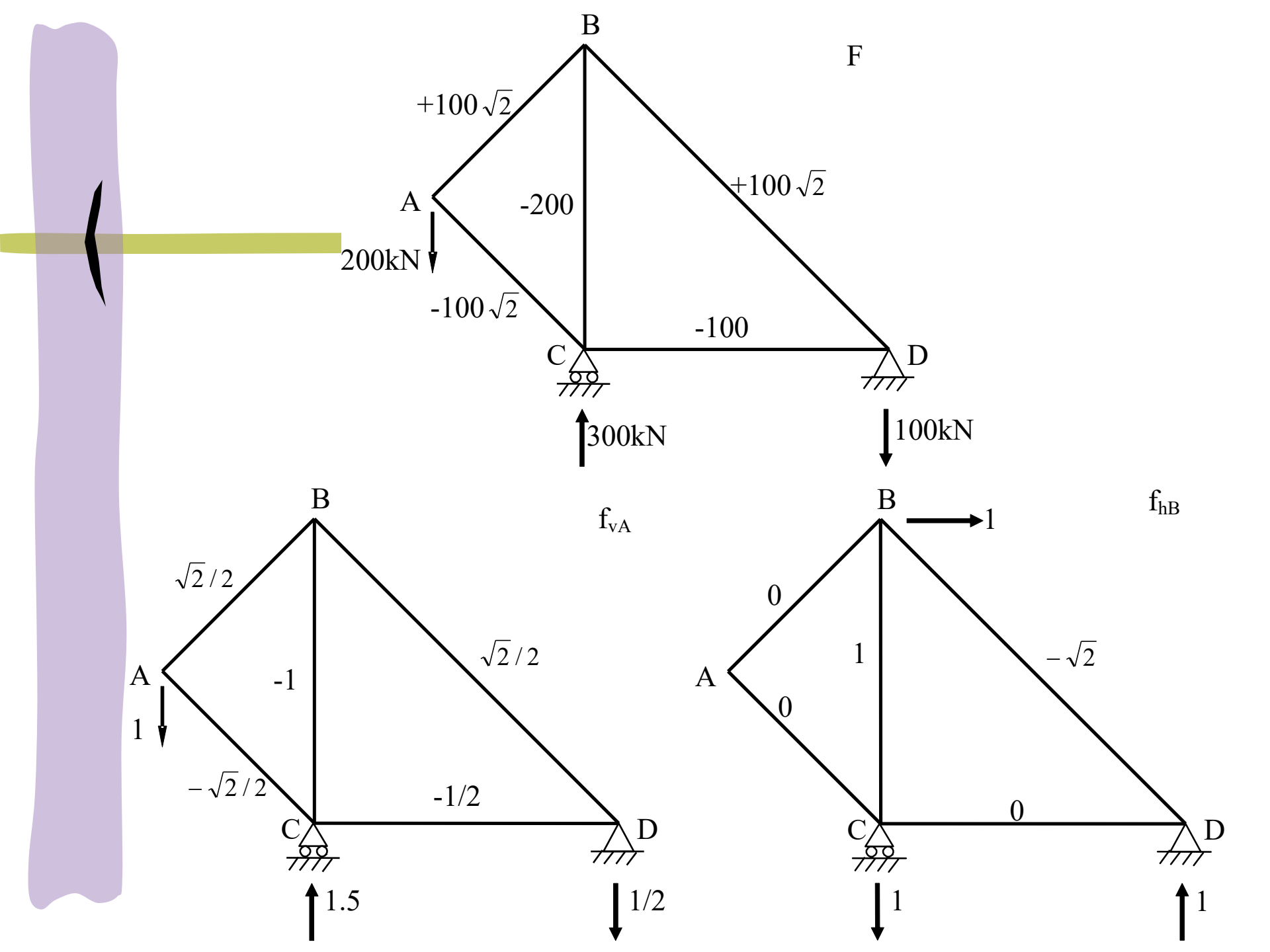
1. Analyze the truss by the method of joints / sections. Find the actual force F in each member
2. Apply a virtual force of one unit at the node and in the direction of the displacement required to be determined. Calculate the internal virtual forces f in all members
3. The actual internal elongation or shortening caused by the actual applied loading in each member = FL/EA
4. Hence $1 \times \delta$ = Sum of $\{f \times FL/EA\}$

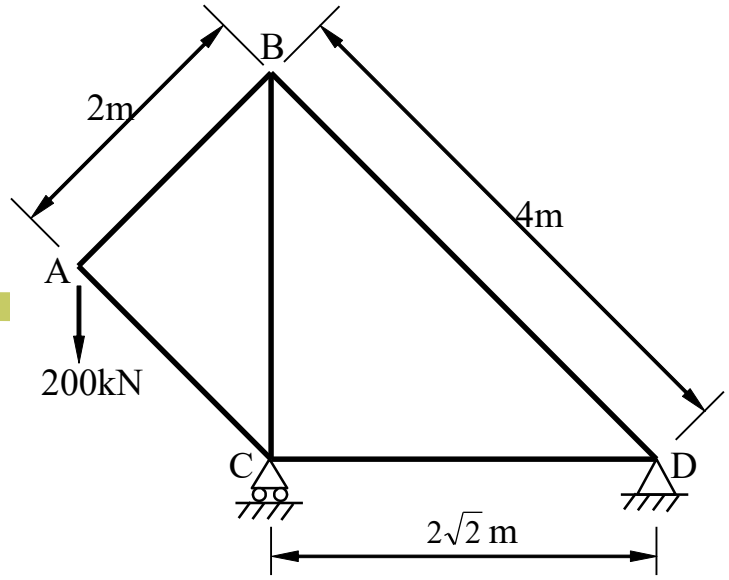
Example 7

Calculate the vertical deflection at A and the horizontal deflection at B. Take $E = 100 \times 10^6$ kN/m^2. The cross sectional areas are given.



A truss under a point load


B
F
+100√2
A
-200
+100√2
200kN
-100√2
-100
C
D
300kN
100kN
B
fvA
√2/2
A
-1
√2/2
1
-√2/2
-1/2
C
D
1.5
1/2
B
1
fhB
0
A
1
-√2
0
0
C
D
1
1

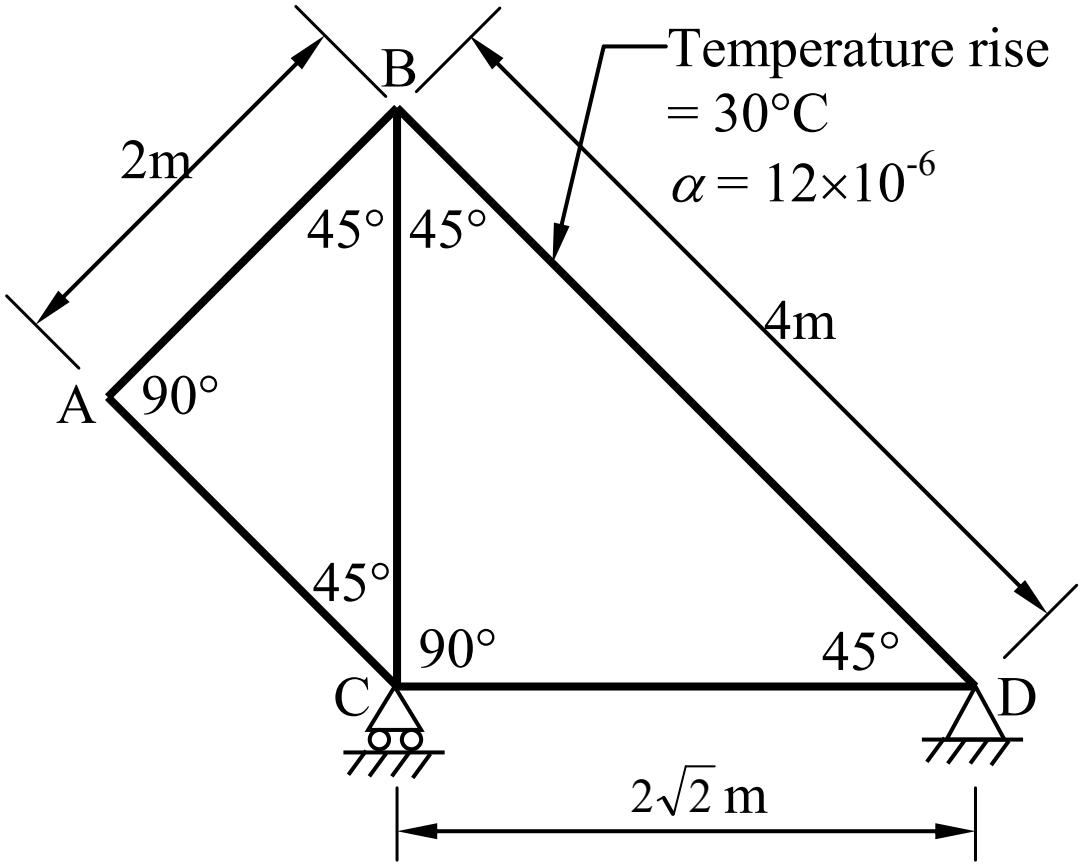


A truss under a point load

Member	Length L (m)	Area A (cm^2)	Force F (kN)	f_{VA}	f_{VA} (FL/A)	f_{HB}	f_{HB} (FL/A)
AB	2	1	$100\sqrt{2}$	$\sqrt{2}/2$	200	0	0
AC	2	1	$-100\sqrt{2}$	$-\sqrt{2}/2$	200	0	0
BC	$2\sqrt{2}$	$\sqrt{2}$	– 200	– 1	400	1	– 400
CD	$2\sqrt{2}$	$\sqrt{2}$	– 100	– 1/2	100	0	0
BD	4	2	$100\sqrt{2}$	$\sqrt{2}/2$	200	$-\sqrt{2}$	– 400
$1\ cm^2 = 0.0001\ m^2$					1100		– 800
$1/E = 1/100\times10^6$					**0.11 m**		**– 0.08 m**

Example 8

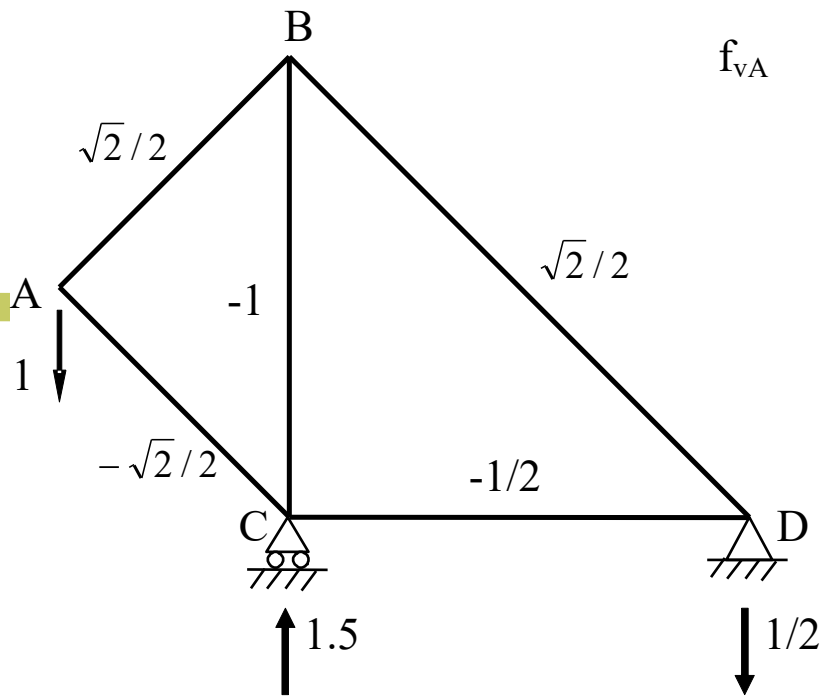
Calculate the vertical deflection at A resulting from a rise in temperature of 30°C in BD. The truss is the same as that in Example 7. Take $\alpha = 12\times10^{-6}$.



A truss under a point load

Deflection of Trusses

1. On a statically determinate truss, no reactions or internal forces can be developed because of a temperature rise or drop t in truss members and hence changes in bar lengths.
2. This will cause the distortion of the whole truss.
3. To find the vertical deflection at A, one may use $1 \times \delta = \text{Sum of } \{f \times \alpha t L\}$
4. Fabrication error or lack of fit can be treated similarly.



Vertical deflection at A

$$1 \times \delta = \text{Sum of } \{f \times \alpha t L\}$$

$$\delta = \sqrt{2}/2 \times (12 \times 10^{-6}) \times 30 \times 4$$

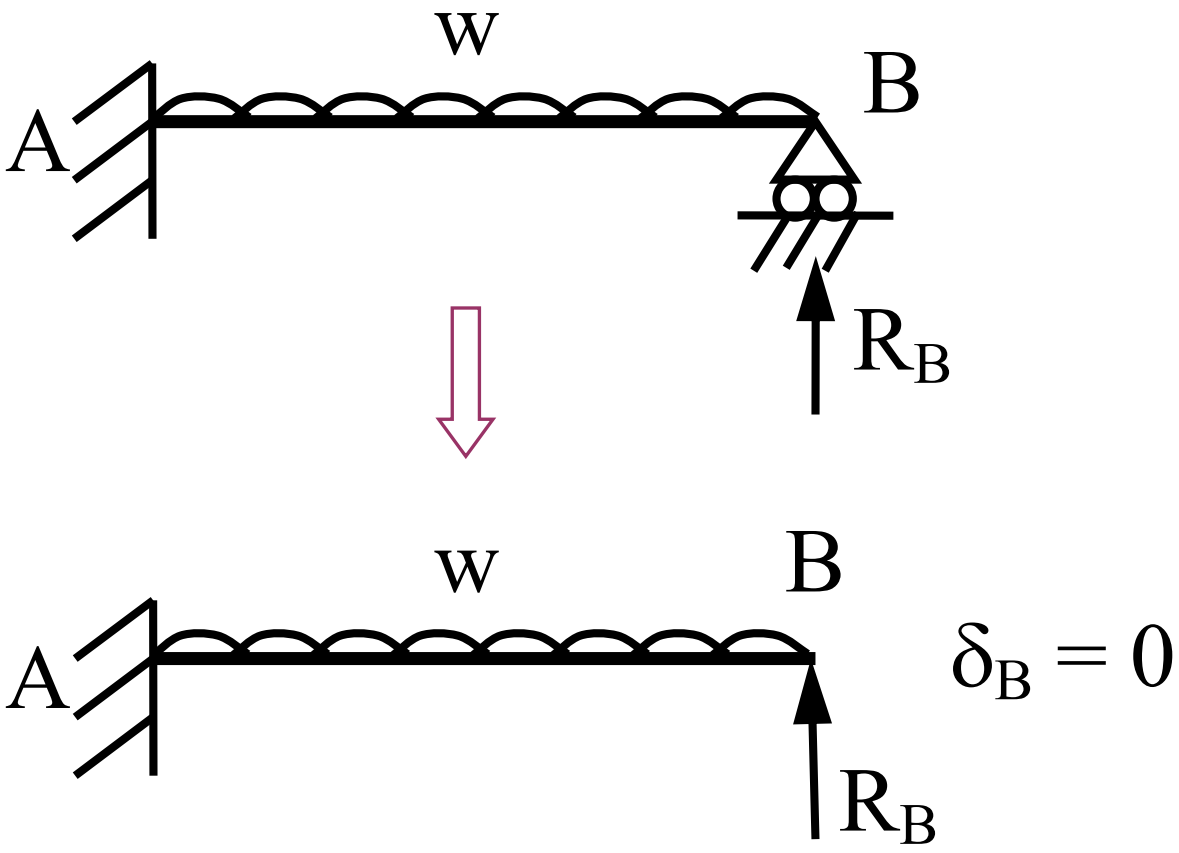
$$= 0.00102 \text{ m}$$

$$= 1.02 \text{ mm}$$

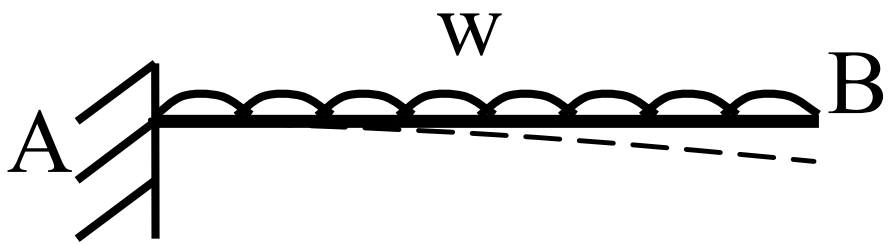
Further Applications of Deflection Calculations



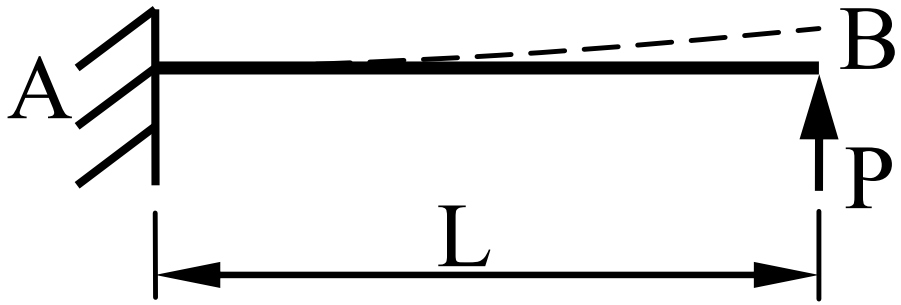
A propped cantilever under UDL
by method of consistent deformation



Cases to consider

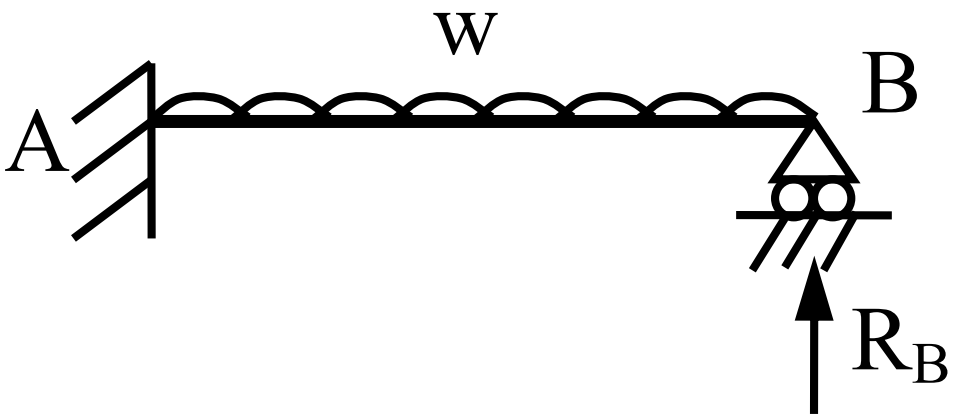


$$\delta_{B1} = \frac{wL^4}{8EI}$$



$$\delta_{B2} = \frac{PL^3}{3EI}$$

A propped cantilever under UDL



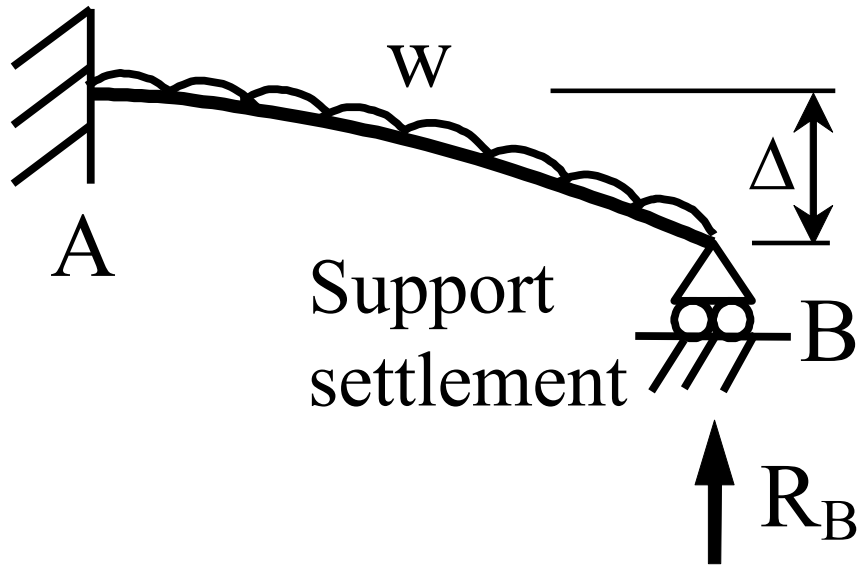
$$\delta_B = \delta_{B1} - \delta_{B2} = 0$$

$$\frac{wL^4}{8EI} - \frac{R_B L^3}{3EI} = 0$$

$$\Rightarrow R_B = \frac{3wL}{8}$$

$$\Rightarrow M$$

A propped cantilever with support settlement



$$\delta_B = \delta_{B1} - \delta_{B2} = \Delta$$

$$\frac{wL^4}{8EI} - \frac{R_B L^3}{3EI} = \Delta$$

$$\Rightarrow R_B = \frac{3wL}{8} - \frac{3EI\Delta}{L^3}$$

$$\Rightarrow M$$

The End

